

MEITECH INTERNATIONAL

Z2 Business, Saif Zone Main Gate, Sharjah, UAE
Phone: +971 6 535 6244 | www.meitech.com

Third Party Subject Matter Expert Assessment

MACHINE SAFEGUARDING EVALUATION REPORT

Sidel Can Line – ISO 12100 / EN ISO 13849-1

Machines + Conveyor Zones



Client:	Coca-Cola Al Ahlia Beverages Company LLC
Facility:	Al Ain Facility, UAE
Project:	Sidel Can Line – Consolidated MSE Assessment
Machines Covered:	Depalletizer, Palletizer, Filler, Can Seamer, OCME Packer + Conveyor Zones
Report No.:	MSE-2026-001
Site Visit:	27th & 28th April 2026
Revision Period:	30 April 2026 to 11 th May
Final Release:	Revision 2.0 – Release
Classification:	CONFIDENTIAL

*Prepared by an Independent Subject Matter Expert on Machine Safety
in accordance with ISO 12100:2010, EN ISO 13849-1:2023, and KORE requirements*

Table of Contents

1. Executive Summary.....	4
2. Project Information.....	5
2.1 Project Timeline	5
2.2 Client Contacts.....	5
2.3 Assessor	5
2.4 Revision History	5
3. Scope & Methodology	6
3.1 Scope	6
3.2 Methodology.....	6
3.3 Limitations	6
4. Standards & Regulations.....	7
4.1 Directives	7
4.2 Standards.....	7
A. Conveyor Zones.....	8
A.1 Action Summary	8
A.2 Scope.....	8
A.3 Risk Estimation	8
A.4 Hazard Zones	9
A.5 KORE Compliance – Conveyor Zones	12
B. Depalletizer	13
B.1 Machine Identification	13
B.2 Action Summary	14
B.3 Overview & Process	14
B.4 Operating Modes	15
B.5 Limits of the Machine.....	15
B.6 Tasks	15
B.7 Risk Summary	16
B.8 Hazard Zones	16
B.9 KORE Compliance – Depalletizer	25
C. Palletizer	26
C.1 Machine Identification.....	26
C.2 Action Summary	27
C.3 Overview & Process	27
C.4 Operating Modes	28
C.5 Limits of the Machine	28
C.6 Tasks	28
C.7 Risk Summary	29
C.8 Hazard Zones	29
C.9 KORE Compliance – Palletizer	38

D. Filler	41
D.1 Machine Identification	41
D.2 Action Summary	42
D.3 Overview & Process	42
D.4 Operating Modes	43
D.5 Limits of the Machine	43
D.6 Tasks	43
D.7 Risk Summary	44
D.8 Hazard Zones	44
D.9 KORE Compliance – Filler	51
E. Can Seamer	52
E.1 Machine Identification	52
E.2 Action Summary	53
E.3 Overview & Process	53
E.4 Operating Modes	53
E.5 Limits of the Machine	54
E.6 Tasks	54
E.7 Risk Summary	55
E.8 Hazard Zones	55
E.9 KORE Compliance – Can Seamer	62
F. OCME Packer	63
F.1 Machine Identification	63
F.2 Action Summary	64
F.3 Overview & Process	64
F.4 Operating Modes	65
F.5 Limits of the Machine	65
F.6 Tasks	65
F.7 Risk Summary	66
F.8 Hazard Zones	66
F.9 KORE Compliance – OCME Packer	75
G. Authorisation & Sign-off	76

1. Executive Summary

This is the consolidated machine safety report for the Sidel Can Line at the Coca-Cola Al Ahlia Beverages facility in Al Ain. It covers five production machines plus the conveyors that connect them, evaluated as a single integrated line.

Site visits took place on 27th and 28th April 2026. The first draft was issued on 30th April 2026 and has gone through multiple internal review cycles since. This document is the final revision.

Method: ISO 12100:2010 risk assessment; risk estimation per EN ISO 13849-1:2023 risk graph. Scope is mechanical and electrical hazards across all foreseeable operating modes and human interaction. Other hazards (chemical, noise quantification, ergonomic ratings) are noted where relevant but not assessed in detail.

Across the line, more than 200 hazard-task combinations were assessed. Common findings:

- Machine-specific LOTO procedures need writing and training, particularly for the OCME packer where some panel circuits remain live with the main isolator off.
- Annual earth continuity and insulation resistance testing should be added to the PM schedule for all five machines.
- Quarterly functional testing of E-stops, light curtains, and door interlocks is recommended.
- Repetitive-task SOPs (jam clearing, film roll changeover, seaming roll changeover, format changeover) are partially in place but need consolidation.

Each machine section ends with its own **KORE** compliance table.

2. Project Information

2.1 Project Timeline

Site Visit	27th & 28th April 2026
Initial Draft	30 April 2026
Revision Period	30 April 2026 onwards
Final Revision	Revision 2.0 Final – current release

2.2 Client Contacts

Name	Role	Department	Contact
Mr.Dionisio C Sedigo	EHS Officer	Health & Safety / QEHS	dionisios@cocacola.co.ae
Mr.Tarek Ibrahim	EHS Manager	Health & Safety / QEHS	tarek.ibrahim@cocacola.co.ae

2.3 Assessor

Capacity	Independent Subject Matter Expert on Machine Safety
Name	Muthu Vignesh R
Qualifications	ANSI B11 LMSS, Reg No - AA311265077 CMSE -TUV NORD® Reg No - 445061746842576
Contact	sales@meitech.com / Mobile no - +97165356244

2.4 Revision History

Version	Date	Description	Author
0.1	28/04/2026	Site visit completed; findings recorded	[SME]
0.2	30/04/2026	Initial draft consolidating all five machines	Meitech Team
1.0	02/05/2026	First internal review – content revisions	[SME]
1.1	10/05/2026	Second review – per-machine KORE, hyperlinked TOC	[SME]
2.0 Final	13/05/2026	Final revision – released to client	[SME]

3. Scope & Methodology

3.1 Scope

Five machines plus the conveyor zones between them. Mechanical and electrical hazards across all foreseeable operating modes and human interactions for each machine.

Out of scope: chemical exposure assessment, quantitative noise measurement, ergonomic scoring (these are noted where observed but not measured).

3.2 Methodology

ISO 12100:2010 risk assessment. Risk estimation using the risk graph method from EN ISO 13849-1:2023 Annex A.

Parameter	Symbol	Description
Severity of injury	S	S1 = Slight (reversible). S2 = Serious (irreversible, including death).
Frequency / duration	F	F1 = Rare / short. F2 = Frequent / long.
Possibility of avoiding	P	P1 = Possible under specific conditions. P2 = Scarcely possible.

Risk Graph

Severity	Frequency	Avoidance	PLr	Risk Level
S1	F1	P1	a	Low
S1	F1	P2	b	Low
S1	F2	P1	b	Medium
S1	F2	P2	c	Medium
S2	F1	P1	c	High
S2	F1	P2	d	High
S2	F2	P1	d	Very High
S2	F2	P2	e	Very High

3.3 Limitations

This report is based on what was observed during the site visits on 27 and 28 April 2026 and information given by site personnel. It does not replace a full risk assessment as required by Machinery Directive 2006/42/EC or applicable UAE workplace health and safety legislation while designing the machine

This report does not verify whether the implemented safety functions meet their required Performance Level. PL verification needs detailed analysis of the safety-related control architecture, which is outside this scope.

Acceptance of residual risk after implementation of recommendations remains with the asset owner.

4. Standards & Regulations

4.1 Directives

Reference	Title
2006/42/EC	Machinery Directive
KORE	Coca-Cola Operating Requirements – Occupational Safety & Health Standards

4.2 Standards

Standard	Title
EN ISO 12100:2010	Safety of machinery – General principles for design – Risk assessment and risk reduction
EN ISO 13849-1:2023	Safety-related parts of control systems – Part 1: General principles
EN 60204-1:2018	Electrical equipment of machines – Part 1: General requirements
EN ISO 13850:2015	Emergency stop function – Principles for design
EN ISO 13854:2019	Minimum gaps to avoid crushing of parts of the human body
EN ISO 13855:2010	Positioning of safeguards with respect to approach speeds
EN ISO 13857:2019	Safety distances to prevent hazard zones being reached
EN ISO 14118:2018	Prevention of unexpected start-up
EN ISO 14119:2013	Interlocking devices associated with guards
EN ISO 14120:2015	Guards – General requirements
EN ISO 14122-1:2016	Permanent means of access to machinery
EN ISO 4414:2010	Pneumatic fluid power – General rules
IEC 62046	Application of protective equipment to detect persons
EN 415-3	Safety of packaging machines – Form, fill and seal machines

A. Conveyor Zones

This section covers the conveyors linking the five machines on the line. Conveyors aren't always given the same attention as headline machines, but operators interact with them more often than anything else — clearing jams, cleaning, walking past them. The hazard categories repeat across every section, so they're documented here once and referenced from each machine.

TOP 3 PRIORITIES

- 1. Verify cascade stop logic between sections** — An E-stop on one conveyor must bring upstream and downstream sections to a controlled stop. Test functional cascade quarterly.
- 2. Set up quarterly E-stop functional test programme** — Test pull-cords and push buttons every quarter. Document activation force, travel, and reset behaviour.
- 3. Write section-specific LOTO procedures** — Each conveyor section needs its own LOTO covering the local isolator, residual energy, and standstill verification.

A.1 Action Summary

Consolidated list of actions identified across all conveyor hazard zones.

Zone	Ref.	Action	Status
CV-01	T012	Cascade stop logic between linked conveyor sections	Found OK
CV-01	A013	Section-specific LOTO procedures for conveyor sections	Found OK
CV-02	A013	Apply LOTO before any work on belt return path	Found OK
CV-03	A013	LOTO with thermal cooldown wait before motor work	Found OK
CV-04	A006	Quarterly E-stop functional test programme	Recommended
CV-05	A013	LOTO with standstill verification before drive maintenance	Critical

A.2 Scope

Roller and chain conveyors between Depalletizer, Filler, Can Seamer, OCME Packer, and Palletizer. Belt returns, drive motors, E-stop coverage, and belt/chain drives are all evaluated here.

A.3 Risk Estimation

Severity	Frequency	Avoidance	PLr	Risk Level
S1	F1	P1	a	Low
S1	F1	P2	b	Low
S1	F2	P1	b	Medium
S1	F2	P2	c	Medium
S2	F1	P1	c	High
S2	F1	P2	d	High
S2	F2	P1	d	Very High
S2	F2	P2	e	Very High

A.4 Hazard Zones

Hazard Zone CV-01: Roller & Chain Conveyors

Hazard Source:	Powered roller and chain conveyors	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R005	Tasks:	T01,T02,T03,T04,T05
Risk Parameters:	S2 / F2 / P1 => PLr d		

Maintain interlocked guarding on chain sprockets and side-rail nip points. Verify cascade stop logic — an E-stop on one section must stop upstream and downstream sections together. Test cascade quarterly.



CV-01: Conveyor section with E-stop

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges on conveyor frames; minimum gap per ISO 13854	Existing
Guard	G001	Fixed guards on roller drive ends and chain returns	Existing
Device	T011	E-Stop pull-cords / push buttons every 10 m along the run	Existing
Device	T012	Cascade stop logic between linked sections	Recommended verification
Admin	A013	Section-specific LOTO before any conveyor work	Recommended corrections
Admin	A019	Mandatory hand and foot protection	Existing

► ACTION REQUIRED

- **[T012]** Cascade stop logic between linked sections — *Recommended verification*
- **[A013]** Section-specific LOTO before any conveyor work — *Recommended corrections*

Hazard Zone CV-02: Belt Returns & Idlers

Hazard Source:	Conveyor belts, idlers, and return paths	Consequence:	Entanglement / Crushing
Risk IDs:	R006-R009	Tasks:	T02,T05,T11
Risk Parameters:	S2 / F1 / P1 => PLr c		

Fit side guards on all belt return rollers. Apply LOTO before any work on belt returns or idlers. Belt tracking adjustment must be done with the conveyor stopped and isolated.

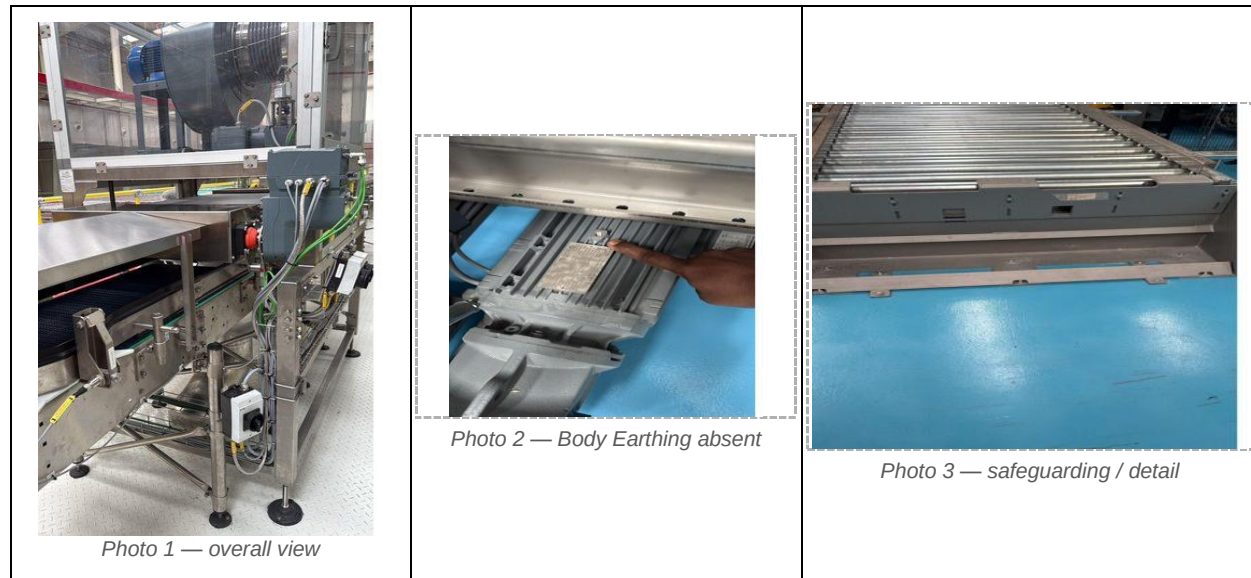


Figure CV-02: Conveyor frame and belt return area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Side guards on belt return rollers	Existing
Guard	G001	Fixed guard on belt drive heads	Existing
Device	T008	Local lockable isolator at each conveyor section	Existing
Admin	A013	LOTO mandatory before any work on belt return path	Critical
Admin	A006	Periodic inspection of belt tracking and idler condition	Existing

► ACTION REQUIRED

- [A013] LOTO mandatory before any work on belt return path — Critical

Hazard Zone CV-03: Drive Motors & Gearboxes

Hazard Source:	Conveyor drive motors and gearboxes	Consequence:	Entanglement / Burn
Risk IDs:	R010-R013	Tasks:	T02,T05,T11
Risk Parameters:	S2 / F1 / P1 => PLr c		

Fit fixed guards on motor couplings and shafts. Display hot surface warnings (motor surfaces can exceed 60°C). Apply LOTO with thermal cooldown before motor work.



Figure CV-03: Conveyor drive motor and gearbox

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Surface temperature limits per ISO 13732-1	Existing
Guard	G001	Fixed guard on motor coupling and shaft	Existing
Device	T008	Local isolator with lockable OFF	Existing
Admin	A013	LOTO with thermal cooldown wait before motor work	Critical
Admin	A002	Hot surface warning on motor enclosures	Existing

► ACTION REQUIRED

- [A013] LOTO with thermal cooldown wait before motor work — Critical

Hazard Zone CV-04: Emergency Stop Coverage

Hazard Source:	E-Stop devices along conveyor runs	Consequence:	Inadequate emergency response
Risk IDs:	R014-R016	Tasks:	T01,T05,T11
Risk Parameters:	S2 / F1 / P1 => PLr c		

Verify every E-stop is reachable from any working position per ISO 13850. Test pull-cords quarterly for activation force and travel — replace any unit out of spec.

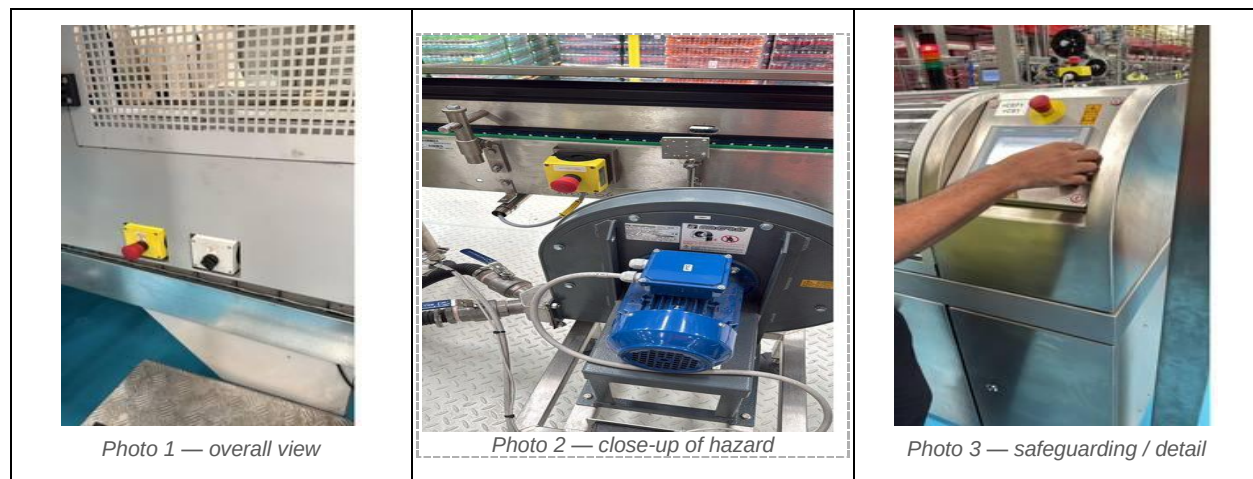


Figure CV-04: E-stop and key selector switch

Risk Reduction Measures:

Category	Ref.	Measure	Status
Device	T011	E-Stop push buttons / pull-cords per ISO 13850	Existing
Admin	A006	Quarterly E-stop functional test	Recommended
Admin	A002	E-stop locations marked with high-visibility signage	Existing

► ACTION REQUIRED

- [A006] Quarterly E-stop functional test — Recommended

A.5 KORE Compliance – Conveyor Zones

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	ISO 12100 risk assessment completed for all conveyor zones.	Compliant	Sec. A.4
Machine Guarding	Fixed guards on drives and belt returns adequate.	Compliant	Sec. A.4
LOTO	Local isolators per section. Section-specific LOTO procedures to be written.	Action Req.	Sec. A.4
Emergency Stop	Coverage adequate. Quarterly functional test programme to be set up.	Action Req.	CV-04
Hot Surface Risk	Warning signs in place on drive motors.	Compliant	CV-03
Inspection & PM	Drive guards, belt tracking, E-stops to be added to PM schedule.	Action Req.	Sec. A.3

B. Depalletizer

First machine on the line. Pulls empty cans off incoming pallets and feeds them into the conveyor system. Eight hazard zones identified during the site visit; details below.

TOP 3 PRIORITIES	
1. Machine-specific LOTO procedure	— Cover electrical and pneumatic isolation; mechanical block engagement on the vertical arm. Train all maintenance staff before next major intervention.
2. Verify body earthing on all motors	— Test continuity on each motor enclosure per IEC 60204-1 Clause 8.2.3. Add second PE conductor where missing. Document each motor.
3. Trapped-key system on access doors	— Replace standard keys with proper trapped-key interlock per ISO 14119. Audit each interlock and procure replacements.

B.1 Machine Identification

Machine Name	Depalletizer
Manufacturer	[refer to nameplate photo]
Machine Family	DEPALLETIZER
Type / Model	Sweep Off
Serial ID	SIBG000300329
Year of Manufacture	2025
Rated Voltage	400 V
Full Load Current	42 A
Control Voltage	24 VDC
CE Marking	Non CE Marked

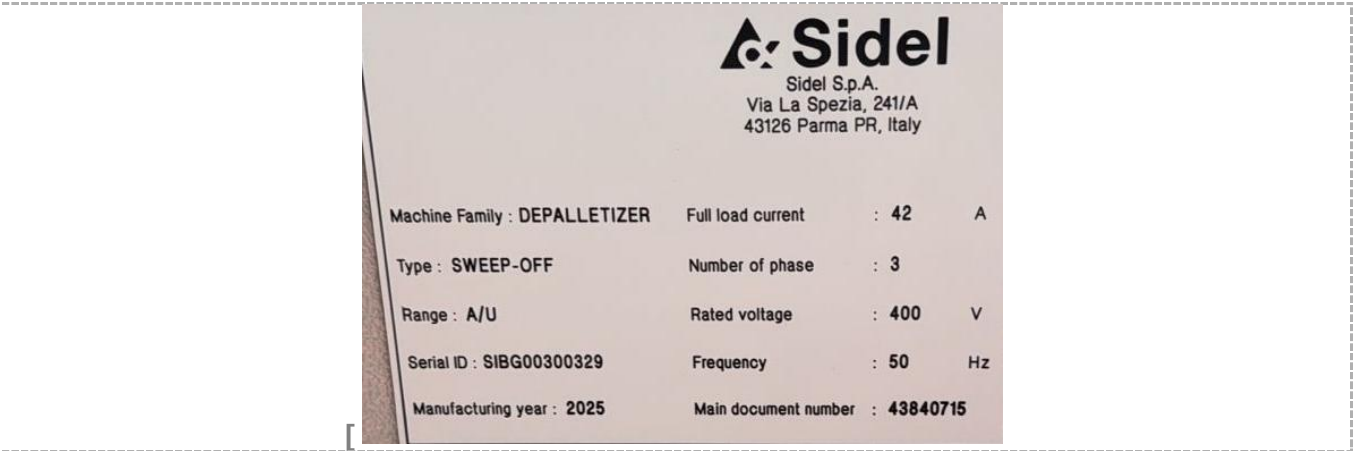


Figure B.1: Depalletizer nameplate

B.2 Action Summary

Consolidated list of actions identified across all hazard zones for the Depalletizer. Detail and supporting context follow in subsequent sections.

Zone	Ref.	Action	Status
HZ-01	A013	LOTO with pneumatic isolation before access	Recommended corrections
HZ-02	A013	LOTO with pneumatic isolation before pusher access	Critical
HZ-03	A013	LOTO + mechanical block engaged before arm access	Critical
HZ-04	A013	Machine-specific LOTO SOP	Critical
HZ-04	A004	Live work by qualified electricians only	Recommended
HZ-05	A013	LOTO with pneumatic isolation and pressure bleed-down	Critical
HZ-06	A013	LOTO before any electrical work	Critical
HZ-06	A006	Annual earth continuity and insulation resistance testing	Critical
HZ-07	A013	LOTO before any work in pusher arm sweep zone	Critical

B.3 Overview & Process

The Depalletizer takes incoming pallets of empty cans and decants them onto the line. It uses pneumatic pushers and clamps, a vertical moving arm, and discharge conveyors. The whole working envelope is enclosed in perimeter guarding with interlocked access doors and safety light curtains.

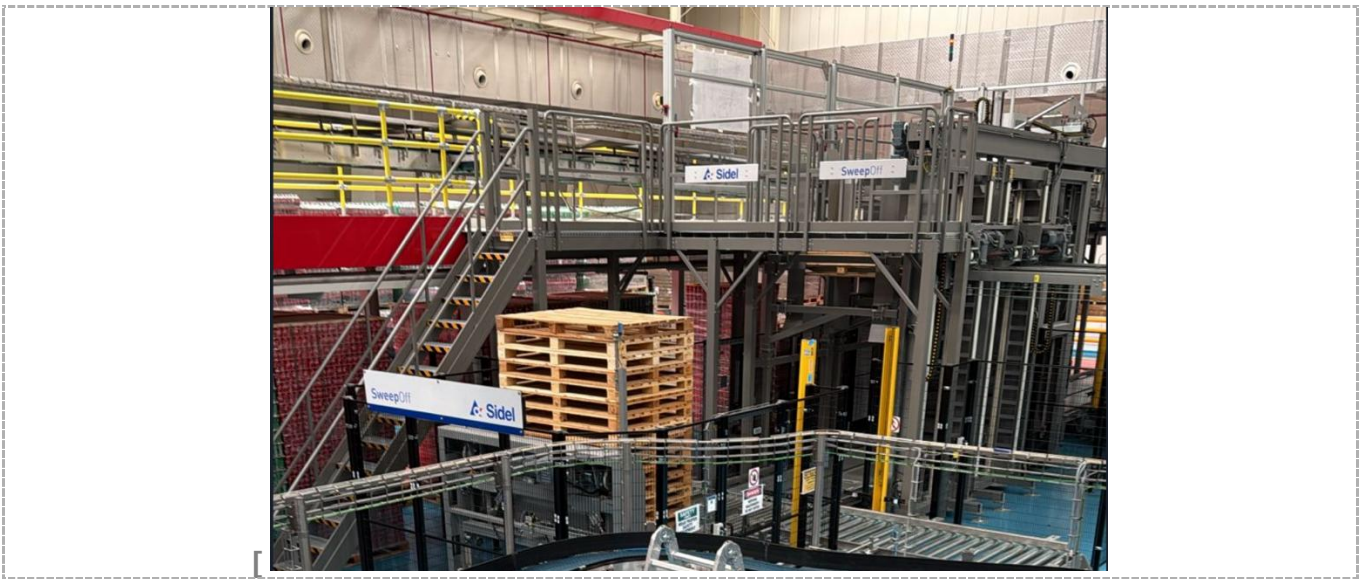


Figure B.2: Depalletizer – machine overview

Pallets enter on the infeed conveyor. Pneumatic clamps secure each one. The pusher transfers can layers from the pallet onto an elevation platform. The vertical arm lifts the layers and positions them onto the line conveyor. Empty pallets discharge to the empty pallet stack.

B.4 Operating Modes

Mode	Description
Automatic	Normal production. PLC controls all motion. Operator only monitors and clears jams from designated points.
Manual / Jog	Reduced-speed jog under operator control. Used for setup, fault-finding, and changeover.

B.5 Limits of the Machine

Intended Use

Removing empty cans from incoming pallets and transferring them onto the can line during normal production.

Foreseeable Misuse

Entering the moving arm zone while in automatic mode. Disabling interlocks or light curtains. Skipping pneumatic isolation during maintenance. Bypassing guard door interlocks. Climbing on the structure.

Known Faults

Pallet misalignment causing pusher jam. Clamp failing to engage. Vertical arm position sensor faults. Control system errors causing unintended movement.

B.6 Tasks

Tasks observed during the site visit. These cover both routine production and maintenance activity.

No.	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor / device adjustment	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans / pallets	Operator	Stopped	As needed
T06	Manual jog of pusher / arm	Operator	Manual	As needed
T07	Inspection of arm and clamp areas	Operator	Auto	Continuous
T08	Pallet stack management	Operator	Auto	Continuous
T09	HMI parameter changes	Maintenance	Manual	Periodic
T10	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T11	Pneumatic system maintenance	Maintenance	LOTO	Scheduled
T12	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T13	Format / changeover	Maintenance	Manual	Periodic

B.7 Risk Summary

Zone	Hazard Source	Consequence	Risk IDs	Tasks	S/F/P	PLr	Achieved
HZ-01	Conveyors, Pushers & Clamps	Crushing	R001-R005	T01-T05	S2/F2/P1	D	Compliant
HZ-02	Pusher Top Area	Crushing	R006-R009	T02,T05,T06	S2/F1/P1	C	Compliant
HZ-03	Vertical Moving Arm	Crushing	R010-R014	T02,T05,T07	S2/F2/P2	E	Compliant
HZ-04	Control Panel / Live Terminals	Electrocution	R015-R019	T02,T09,T11	S2/F1/P2	NA	Compliant
HZ-05	Pneumatic Cylinders & Clamps	Impact/Crushing	R020-R023	T02,T05,T10	S2/F1/P1	C	Compliant
HZ-06	Electrical Earthing	Electrocution	R024-R028	T02,T05,T11	S2/F2/P1	NA	Compliant
HZ-07	Pusher Arm	Crushing	R029-R032	T02,T05,T07	S2/F1/P1	C	Compliant
HZ-08	General Moving Parts	Loss of Awareness	R033-R035	T01,T11,T13	S1/F2/P2	NA	Compliant

B.8 Hazard Zones

Hazard Zone HZ-01: Conveyors, Pushers & Clamps

Hazard Source:	Pallet/can conveyors, pushers, clamping mechanisms	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R005	Tasks:	T01,T02,T03,T04,T05
Risk Parameters:	S2 / F2 / P1 => PLr d		

Apply LOTO with pneumatic isolation before any access. Train operators on safe jam-clearing — pneumatic clamps can release unexpectedly without pressure isolation.

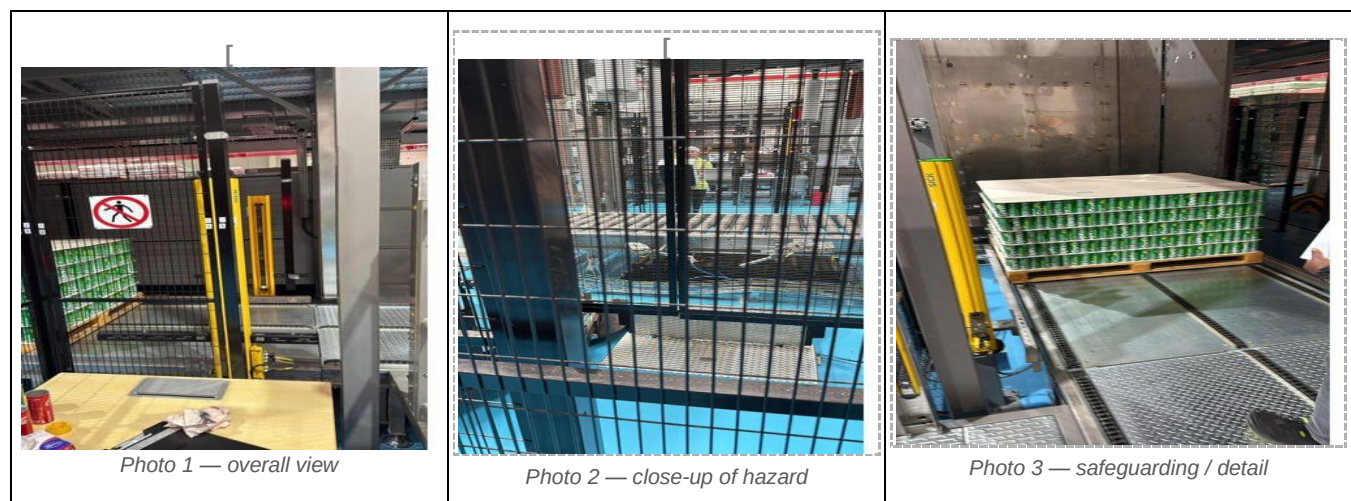


Figure HZ-01: Depalletizer infeed pusher and clamp area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges; minimum gap per ISO 13854	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Device	T010	Safety light curtains at material entry/exit, PLr e	Existing
Device	T014	Interlock with panic release per ISO 14119, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T011	E-Stop push buttons per ISO 13850	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Admin	A013	LOTO with pneumatic isolation before access	Recommended corrections
Admin	A002	Awareness signs at access points	Existing

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation before access — Recommended corrections

Hazard Zone HZ-02: Pusher Top Area

Hazard Source:	Top pusher mechanism and pneumatic actuators	Consequence:	Crushing / Impact
Risk IDs:	R006-R009	Tasks:	T02,T05,T06,T07
Risk Parameters:	S2 / F1 / P1 => PLr c		

Pneumatic pusher transfers can layers from pallet to discharge conveyor. Operates with significant force; contact with the pusher in motion is a serious crush hazard. Apply LOTO with pneumatic isolation and standstill verification before any maintenance work.

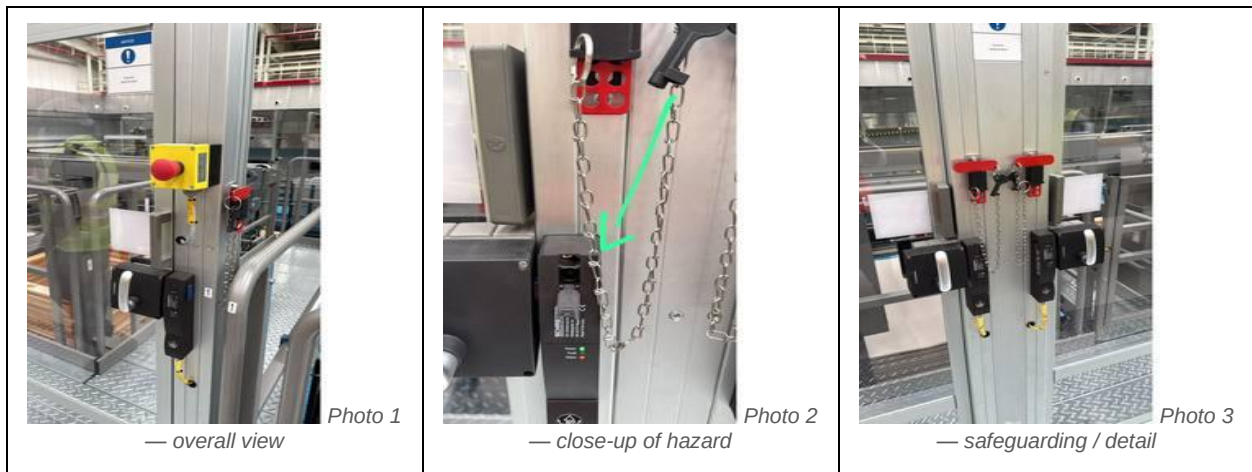


Figure HZ-02: Pusher top area with pneumatic actuators

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical stops on pusher travel	Existing
Guard	G003	Perimeter guard preventing access during automatic mode	Existing
Device	T014	Interlock with panic release on access doors, PLr d	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO with pneumatic isolation before pusher access	Critical
Admin	A019	Mandatory hand and foot protection	Existing

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation before pusher access — Critical

Hazard Zone HZ-03: Vertical Moving Arm/Roller belts

Hazard Source:	Vertical moving arm assembly	Consequence:	Crushing / Impact
Risk IDs:	R010-R014	Tasks:	T02,T05,T07,T10
Risk Parameters:	S2 / F2 / P2 => PLr e		

The arm raises and lowers to lift can layers off the pallet. Treat as a no-go zone during automatic operation. Once movement starts, contact cannot be avoided. Engage mechanical blocks before any access to the sweep zone — verify visually.

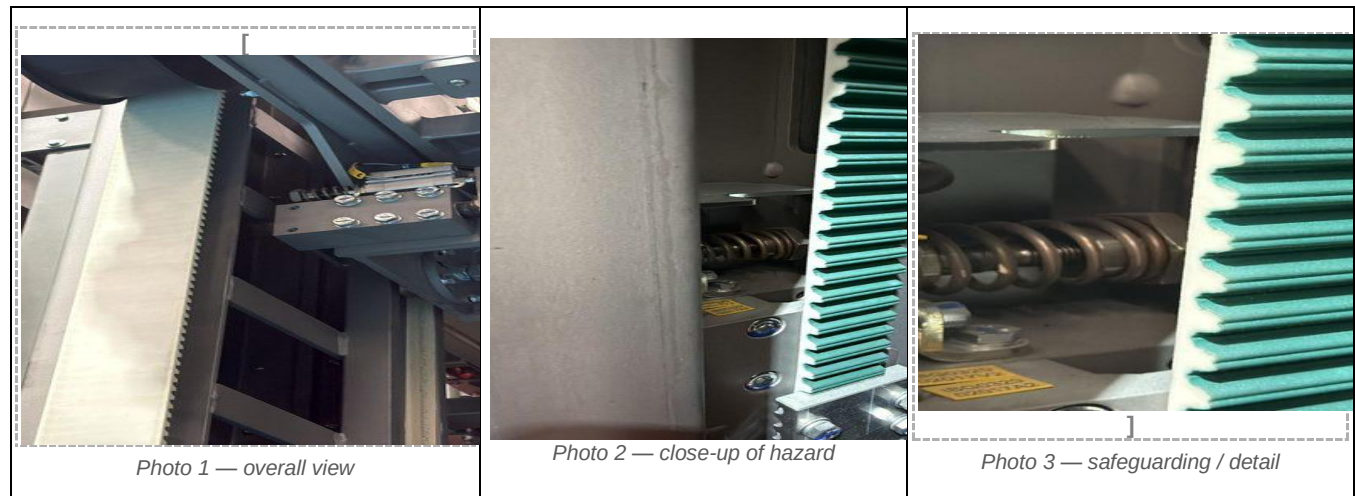


Figure HZ-03: Vertical moving arm

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical blocks for arm at top of stroke	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Device	T010	Safety light curtain, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T005	Reset button inside safeguarded space, PLr e	Existing
Device	T011	E-Stop push buttons	Existing
Device	T015	Mode selector for manual mode, PLr c	Existing
Admin	A013	LOTO + mechanical block engaged before arm access	Critical
Admin	A006	Periodic inspection of mechanical blocks and interlocks	Existing

► ACTION REQUIRED

- **[A013]** LOTO + mechanical block engaged before arm access — Critical

Hazard Zone HZ-04: Control Panel & Live Terminals

Hazard Source:	Main electrical control panel	Consequence:	Electrocution
Risk IDs:	R015-R019	Tasks:	T02,T09,T11
Risk Parameters:	S2 / F1 / P2 => No engineering control Fixed guards is the control measure		

400V terminals inside the main panel. Risk is highest during fault-finding when the panel is open. Identify all permanently live circuits in the machine-specific LOTO procedure. Verify panel warning labels are in place and legible.

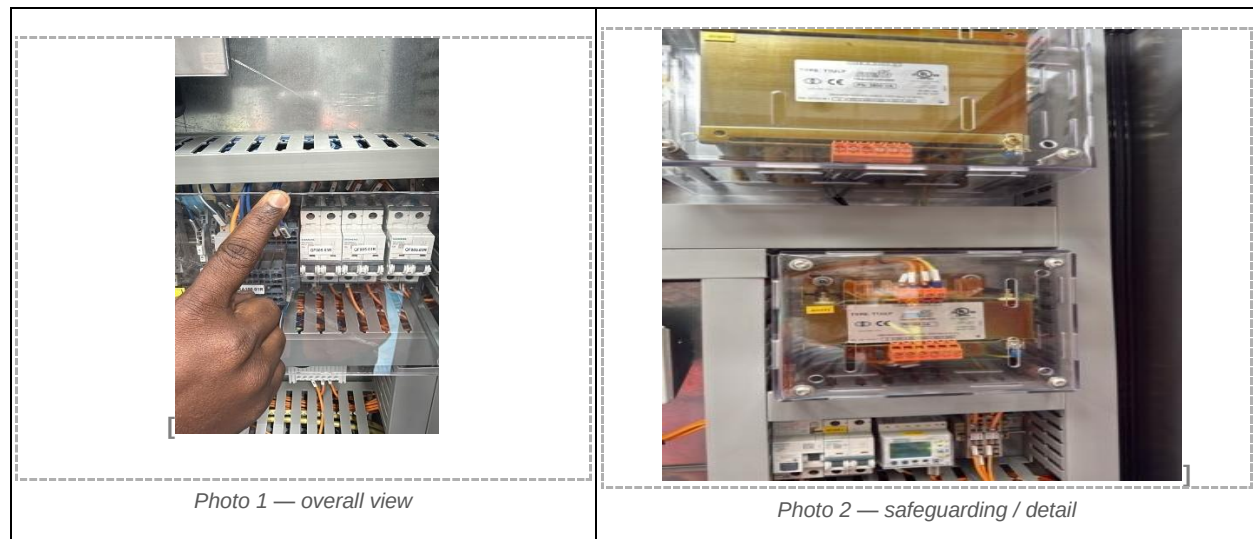


Figure HZ-04: Main electrical panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	IP-rated enclosures	Existing
Guard	G001	Fixed guard on control panel	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T021	Dual earthing PE protection	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	Machine-specific LOTO SOP	Critical
Admin	A004	Live work by qualified electricians only	Recommended
Admin	A002	Electrical hazard signs on panels	Existing

► ACTION REQUIRED

- [A013] Machine-specific LOTO SOP — *Critical*
- [A004] Live work by qualified electricians only — *Recommended*

Hazard Zone HZ-05: Pneumatic Cylinders & Clamps

Hazard Source:	Pneumatic cylinders and clamping devices	Consequence:	Impact / Crushing / Whip
Risk IDs:	R020-R023	Tasks:	T02,T05,T10
Risk Parameters:	S2 / F1 / P1 => PLr c		

Pneumatic energy doesn't disappear when you cut electrical power. Cylinders can move on de-energisation if pressure isn't bled. A loose pressurised hose can whip badly. Include pneumatic isolation and bleed-down as mandatory steps in the LOTO procedure.

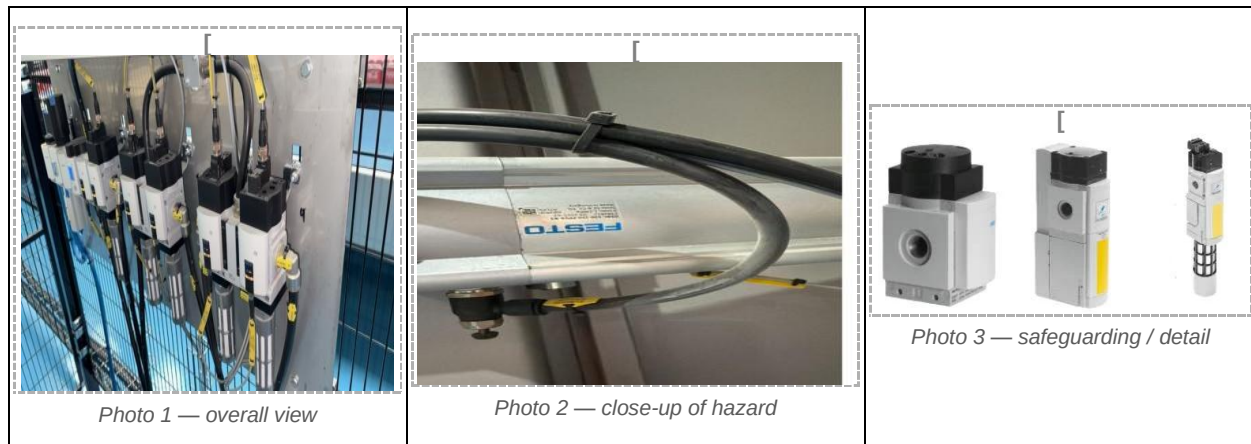


Figure HZ-05: Pneumatic cylinder and pressure regulation

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Shrouded pneumatic connections, secured hose routing	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Device	T011	E-Stop also isolates pneumatic energy	Existing
Admin	A013	LOTO with pneumatic isolation and pressure bleed-down	Critical
Admin	A004	Procedures for jam clearing near pneumatic actuators	Existing

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation and pressure bleed-down — *Critical*

Hazard Zone HZ-06: Electrical Earthing & Over-Voltage

Hazard Source:	Machine frame earthing and over-voltage protection	Consequence:	Electrocution / Shock
Risk IDs:	R024-R028	Tasks:	T02,T05,T11
Risk Parameters:	S2 / F2 / P1 => PLr d		

IEC 60204-1 Clause 8.2.3 requires two independent protective conductors. A single earth path failing can energise the frame. Schedule annual earth continuity and insulation resistance testing. Document each test in the PM register.

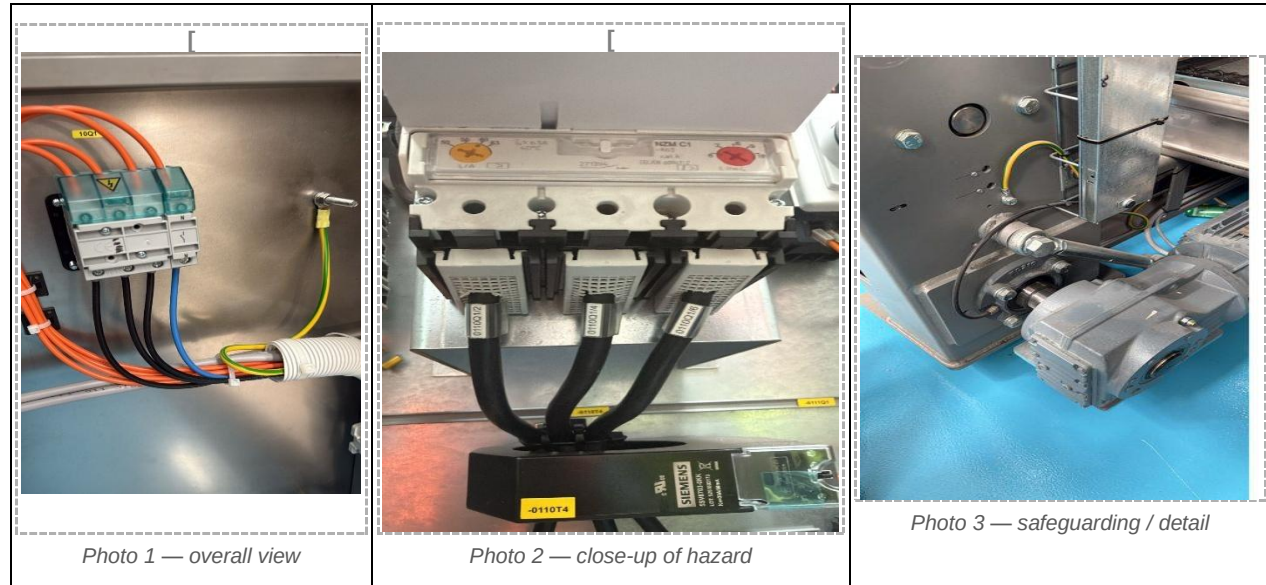


Figure HZ-06: Earthing and surge protection in panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Equipotential bonding of all metalwork	Existing – body earthing to be verified
Device	T021	Dual earthing PE protection	Existing
Device	T019	Surge protection device	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Admin	A013	LOTO before any electrical work	Critical
Admin	A006	Annual earth continuity and insulation resistance testing	Critical

► ACTION REQUIRED

- [A013] LOTO before any electrical work — Critical
- [A006] Annual earth continuity and insulation resistance testing — Critical

Hazard Zone HZ-07: Pusher Arm

Hazard Source:	Powered pusher arm assembly	Consequence:	Crushing / Impact
Risk IDs:	R029-R032	Tasks:	T02,T05,T07
Risk Parameters:	S2 / F1 / P1 => PLr c		

Maintain interlocked guarding around pusher arm sweep zone. Verify interlock function quarterly. Apply LOTO before any access.



Photo 1 — overall view

Figure HZ-07: Pusher arm and drive mechanism

Risk Reduction Measures:

Category	Ref.	Measure	Status
Guard	G003	Perimeter guard with interlocked access	Existing
Device	T014	Interlock with panic release	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO before any work in pusher arm sweep zone	Critical
Admin	A002	Warning signs at pusher arm access points	Existing

► ACTION REQUIRED

- [A013] LOTO before any work in pusher arm sweep zone — Critical

Hazard Zone HZ-08: General Moving Parts – Awareness

Hazard Source:	Multiple simultaneously moving assemblies	Consequence:	Loss of Awareness / Ergonomic
Risk IDs:	R033-R035	Tasks:	T01,T11,T13
Risk Parameters:	S1 / F2 / P2 => PLr c		

Reinforce hazard awareness through visible stack lights and pre-start audible warnings. Rotate operators on repetitive tasks.



Photo 1 — overall view

Figure HZ-08: General machine area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Transparent guard panels where practical	Existing
Device	T001	Audible/visible warning before automatic start	Existing
Admin	A001	Stack lights/beacons during operation	Existing
Admin	A002	Awareness signs at access points	Existing
Admin	A019	Mandatory PPE in machine area	Existing

B.9 KORE Compliance – Depalletizer

KORE compliance status for the Depalletizer. Items marked Action Req. require client follow-up.

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	Full ISO 12100 assessment completed; 35+ hazard-task combinations.	Compliant	Sec. B.7
Machine Guarding	Perimeter guards, interlocked doors, light curtains adequate.	Compliant	B.7
LOTO	Electrical and pneumatic energy isolation procedures to be machine-specific.	Action Req.	HZ-04, HZ-05
Electrical Safety	Main isolator, E-stops, dual earthing PE in place.	Compliant	HZ-04, HZ-06
Emergency Stop	Coverage adequate at operator stations.	Compliant	B.7
Training & Competency	Task-specific SOPs to be issued and trained.	Action Req.	B.4, B.5
Inspection & PM	Periodic inspection programme to be set up.	Action Req.	B.7
Contractor & Visitor Safety	Visitor exposure assessed; signage in place.	Compliant	HZ-01
PPE	Safety footwear and hand protection mandatory.	Compliant	B.7
Management of Change	Modifications require re-assessment.	Advisory	-

C. Palletizer

Final machine on the line. Stacks finished can packs onto pallets ready for warehouse dispatch. Eight hazard zones; details below.

TOP 3 PRIORITIES

- 1. Machine-specific LOTO procedure** — Cover all energy sources including mechanical block engagement on the vertical arm. Train all maintenance staff.
- 2. Annual earth continuity test programme** — Schedule and document annual earth continuity and insulation resistance testing for all panels and motors.
- 3. Changeover SOP — manual mode + reduced-speed jog** — Issue written changeover procedure covering guide adjusters and tool lifter. Mandatory mode selector position before adjustment.

C.1 Machine Identification

Machine Name	Palletizer
Manufacturer	[refer to nameplate photo]
Machine Family	PALLETIZER
Type / Model	EVOFLEX H.S
Serial ID	SIFU00300103
Year of Manufacture	2023
Rated Voltage	400 V
Full Load Current	85 A
Control Voltage	24 VDC
CE Marking	Non CE Marked

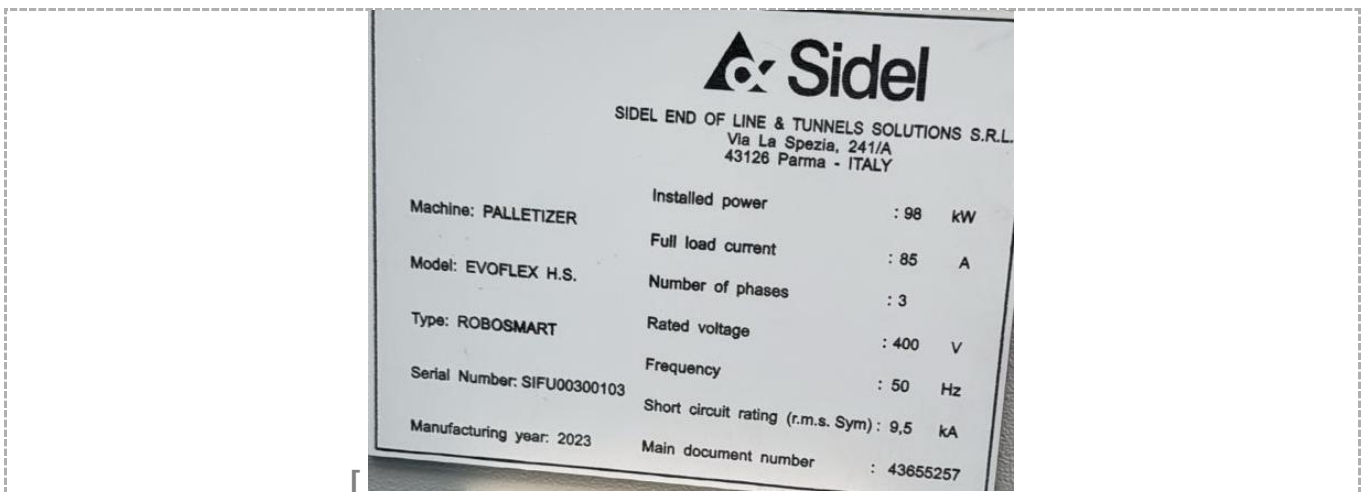


Figure C.1: Palletizer nameplate

C.2 Action Summary

Consolidated list of actions identified across all hazard zones for the Palletizer. Detail and supporting context follow in subsequent sections.

Zone	Ref.	Action	Status
HZ-01	A013	LOTO with pneumatic isolation	Recommended corrections
HZ-02	A013	LOTO before interlayer arm access	Critical
HZ-03	A013	LOTO + mechanical block engaged before arm access	Critical
HZ-04	A013	Machine-specific LOTO SOP	Critical
HZ-04	A004	Live work by qualified electricians only	Recommended
HZ-05	A013	LOTO with pneumatic isolation and bleed-down	Critical
HZ-06	A013	LOTO before any electrical work	Critical
HZ-06	A006	Annual earth continuity / insulation resistance testing	Critical
HZ-07	A013	LOTO before guide adjuster maintenance	Critical
HZ-07	A004	Written changeover SOP specifying manual mode	Recommended

C.3 Overview & Process

The Palletizer takes finished can packs from the line and stacks them onto pallets in defined patterns. It uses an interlayer pad placement arm, a vertical moving arm for layer placement, and pallet discharge. Working envelope enclosed by perimeter guarding with interlocked access doors and safety light curtains.



Figure C.2: Palletizer – machine overview

Finished packs arrive on the infeed conveyor and are positioned by guides into a layer formation. The vertical arm transfers each completed layer onto the building pallet. Between layers, the interlayer arm places a separator pad. Once the pallet reaches programmed height, it discharges to the wrapping or warehouse station.

C.4 Operating Modes

Mode	Description
Automatic	Normal production. PLC controls all motion. Operator only monitors and clears jams from designated points.
Manual / Jog	Reduced-speed jog under operator control. Used for setup, fault-finding, and changeover.

C.5 Limits of the Machine

Intended Use

Stacking finished can packs onto pallets in defined layer patterns during normal production.

Foreseeable Misuse

Entering the moving arm zone while in automatic mode. Disabling interlocks or light curtains. Skipping LOTO. Bypassing guard door interlocks. Manual guide adjustment with the machine running.

Known Faults

Layer pattern misalignment causing arm jam. Interlayer pad misfeed. Pallet height sensor faults. Control system errors causing unintended movement.

C.6 Tasks

Tasks observed during the site visit. These cover both routine production and maintenance activity.

No.	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor / device adjustment	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed packs / pads	Operator	Stopped	As needed
T06	Manual jog of arms	Operator	Manual	As needed
T07	Inspection of arm and pad areas	Operator	Auto	Continuous
T08	Pallet / pad replenishment	Operator	Auto	Continuous
T09	HMI parameter changes	Maintenance	Manual	Periodic
T10	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T11	Pneumatic / drive maintenance	Maintenance	LOTO	Scheduled
T12	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T13	Format / changeover	Maintenance	Manual	Periodic

C.7 Risk Summary

Zone	Hazard Source	Consequence	Risk IDs	Tasks	S/F/P	PLr	Achieved
HZ-01	Conveyors, Pushers & Clamps	Crushing	R001-R005	T01-T05	S2/F2/P1	D	Compliant
HZ-02	Interlayer Arm	Crushing	R006-R009	T02,T05,T06	S2/F1/P1	C	Compliant
HZ-03	Vertical Moving Arm	Crushing	R010-R014	T02,T05,T07	S2/F2/P2	E	Compliant
HZ-04	Control Panel / Live Terminals	Electrocution	R015-R019	T02,T09,T11	S2/F1/P2	NA	Compliant
HZ-05	Pneumatic Cylinders & Clamps	Impact/Crushing	R020-R023	T02,T05,T10	S2/F1/P1	C	Compliant
HZ-06	Electrical Earthing	Electrocution	R024-R028	T02,T05,T11	S2/F2/P1	NA	Compliant
HZ-07	Guide Adjusters	Crushing	R029-R032	T02,T11,T13	S2/F1/P1	C	Compliant
HZ-08	General Moving Parts	Loss of Awareness	R033-R035	T01,T11,T13	S1/F2/P2	NA	Compliant

C.8 Hazard Zones

Hazard Zone HZ-01: Conveyors, Pushers & Clamps

Hazard Source:	Pallet/can conveyors, pushers, clamping mechanisms	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R005	Tasks:	T01,T02,T03,T04,T05
Risk Parameters:	S2 / F2 / P1 => PLr d		

Apply LOTO with pneumatic isolation before access to nip points. Train operators on safe approach during jam-clearing.

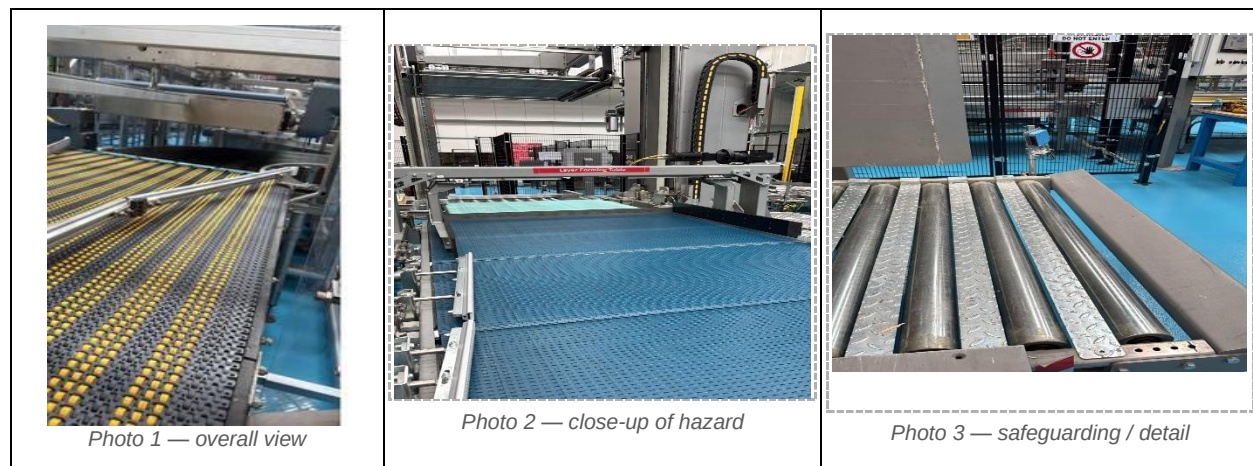


Figure HZ-01: Palletizer infed conveyor

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges; minimum gap per ISO 13854	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Device	T010	Safety light curtains at material entry/exit, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T011	E-Stop push buttons	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Admin	A013	LOTO with pneumatic isolation	Recommended corrections
Admin	A002	Awareness signs at access points	Existing

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation — Recommended corrections

Hazard Zone HZ-02: Interlayer Arm

Hazard Source:	Interlayer pad placement arm	Consequence:	Crushing / Impact
Risk IDs:	R006-R009	Tasks:	T02,T05,T06,T07
Risk Parameters:	S2 / F1 / P1 => PLr c		

Treat interlayer arm zone as no-go during operation. Apply LOTO before access.

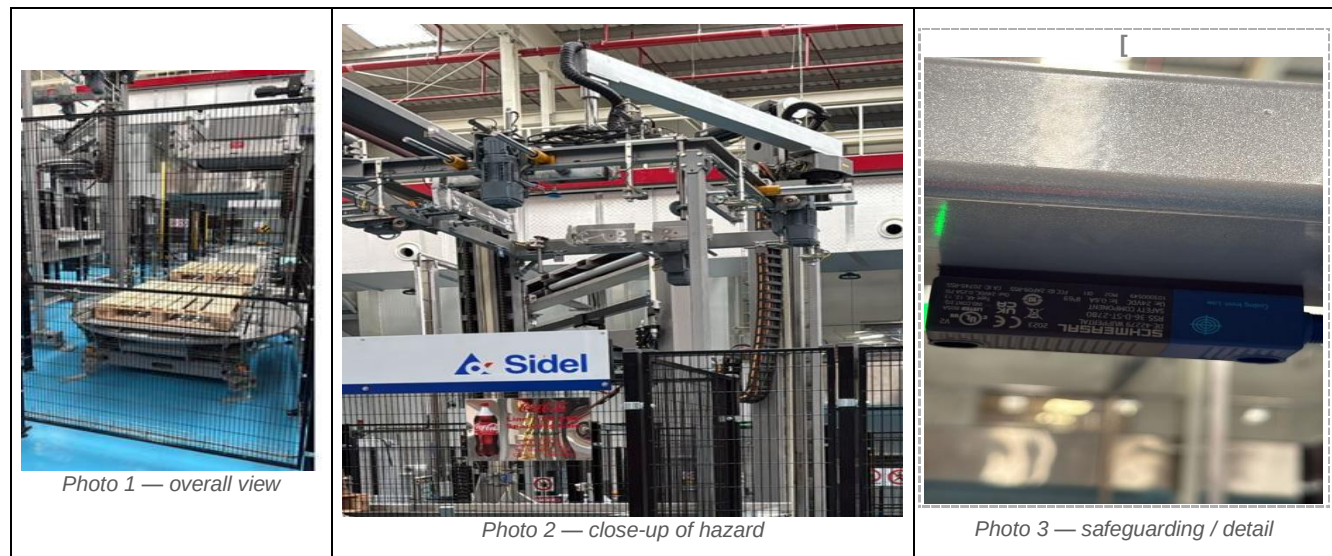


Figure HZ-02: Interlayer arm and pad placement

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical stops on arm travel	Existing
Guard	G003	Perimeter guard preventing access during operation	Existing
Device	T014	Interlock with panic release	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO before interlayer arm access	Critical
Admin	A019	Mandatory hand and foot protection	Existing

► ACTION REQUIRED

- [A013] LOTO before interlayer arm access — Critical

Hazard Zone HZ-03: Vertical Moving Arm

Hazard Source:	Vertical moving arm for layer placement	Consequence:	Crushing / Impact
Risk IDs:	R010-R014	Tasks:	T02,T05,T07,T10
Risk Parameters:	S2 / F2 / P2 => PLr e		

Engage mechanical blocks before any access to the arm sweep zone — verify visually. Apply LOTO with standstill verification.

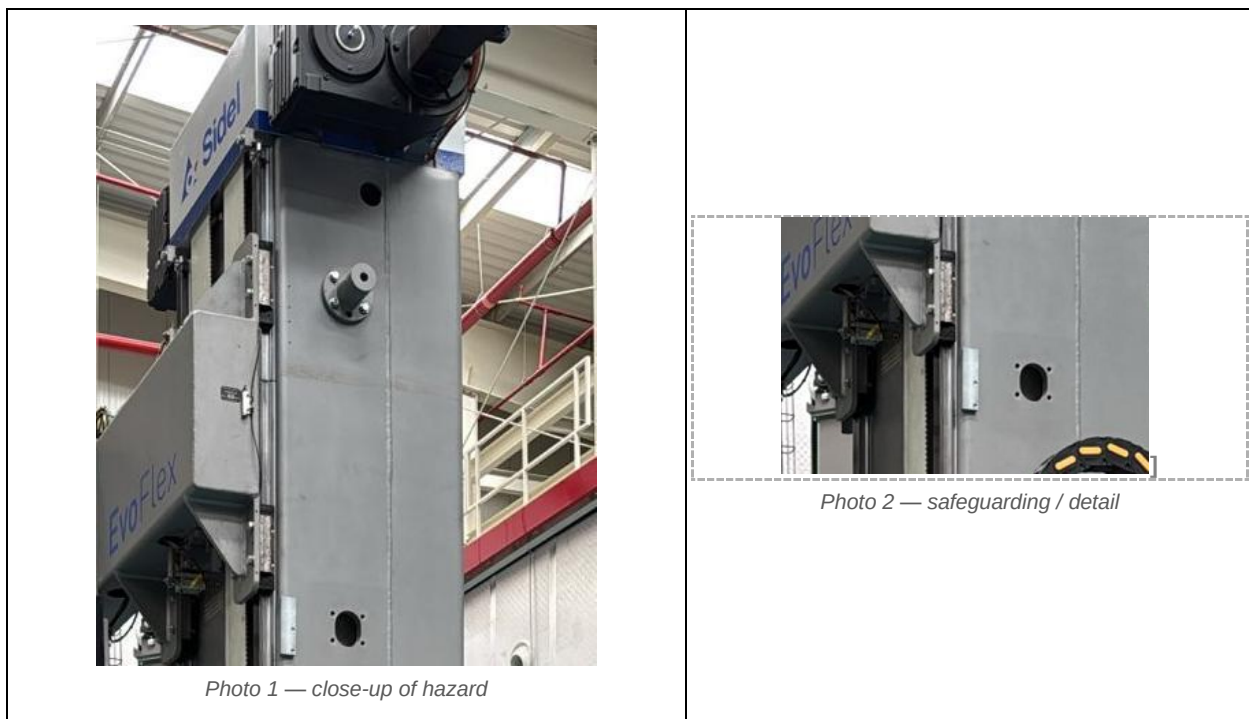


Figure HZ-03: Vertical moving arm

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical blocks for arm at top of stroke	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Device	T010	Safety light curtain, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing key in pocket recommended to update
Device	T005	Reset button inside safeguarded space, PLr e	Existing
Device	T011	E-Stop push buttons	Existing
Device	T015	Mode selector for manual mode, PLr c	Existing
Admin	A013	LOTO + mechanical block engaged before arm access	Critical

► ACTION REQUIRED

- [A013] LOTO + mechanical block engaged before arm access — Critical

Hazard Zone HZ-04: Control Panel & Live Terminals

Hazard Source:	Main electrical control panel	Consequence:	Electrocution
Risk IDs:	R015-R019	Tasks:	T02,T09,T11
Risk Parameters:	S2 / F1 / P2 => PLr d		

400V terminals inside the main panel. Risk highest during fault-finding with the panel open. Permanently live circuits to be identified in the LOTO procedure.



Figure HZ-04: Main electrical panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	IP-rated enclosures	Existing
Guard	G001	Fixed guard on control panel	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T021	Dual earthing PE protection	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	Machine-specific LOTO SOP	Critical
Admin	A004	Live work by qualified electricians only	Recommended

► ACTION REQUIRED

- **[A013]** Machine-specific LOTO SOP — *Critical*
- **[A004]** Live work by qualified electricians only — *Recommended*

Hazard Zone HZ-05: Pneumatic Cylinders & Clamps

Hazard Source:	Pneumatic cylinders and clamping devices	Consequence:	Impact / Crushing / Whip
Risk IDs:	R020-R023	Tasks:	T02,T05,T10
Risk Parameters:	S2 / F1 / P1 => PLr c		

Include pneumatic shut-off and bleed-down in the LOTO procedure. Secure all pressurised hoses to prevent whip on disconnection.



Figure HZ-05: Pneumatic valve island and regulators

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Shrouded pneumatic connections	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Device	T011	E-Stop also isolates pneumatic energy	Existing
Admin	A013	LOTO with pneumatic isolation and bleed-down	Critical

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation and bleed-down — Critical

Hazard Zone HZ-06: Electrical Earthing & Over-Voltage

Hazard Source:	Machine frame earthing and over-voltage protection	Consequence:	Electrocution / Shock
Risk IDs:	R024-R028	Tasks:	T02,T05,T11
Risk Parameters:	S2 / F2 / P1 => PLr d		

Install two independent protective conductors per IEC 60204-1. Schedule annual continuity and insulation resistance testing.

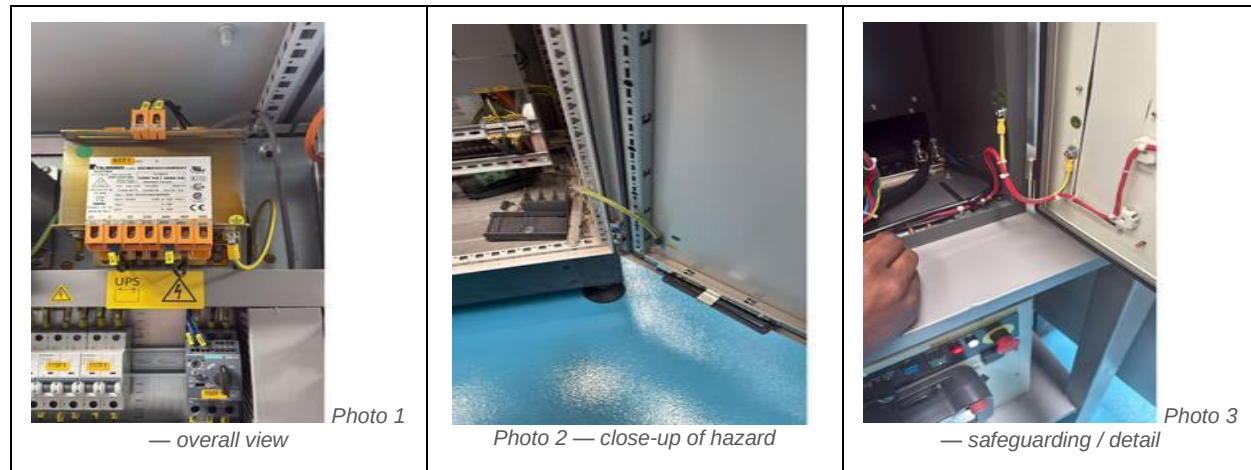


Figure HZ-06: Surge protection and earthing in main panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Equipotential bonding of all metalwork	Existing – body earthing to be verified
Device	T021	Dual earthing PE protection	Existing
Device	T019	Surge protection device	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Admin	A013	LOTO before any electrical work	Critical
Admin	A006	Annual earth continuity / insulation resistance testing	Critical

► ACTION REQUIRED

- [A013] LOTO before any electrical work — Critical
- [A006] Annual earth continuity / insulation resistance testing — Critical

Hazard Zone HZ-07: Guide Adjusters / Tool Lifter

Hazard Source:	Adjustable guides and tool lifting mechanisms	Consequence:	Crushing / Pinch
Risk IDs:	R029-R032	Tasks:	T02,T11,T13
Risk Parameters:	S2 / F1 / P1 => PLr c		

Switch to manual mode and reduced-speed jog before format changeover. Document this in a written changeover SOP.

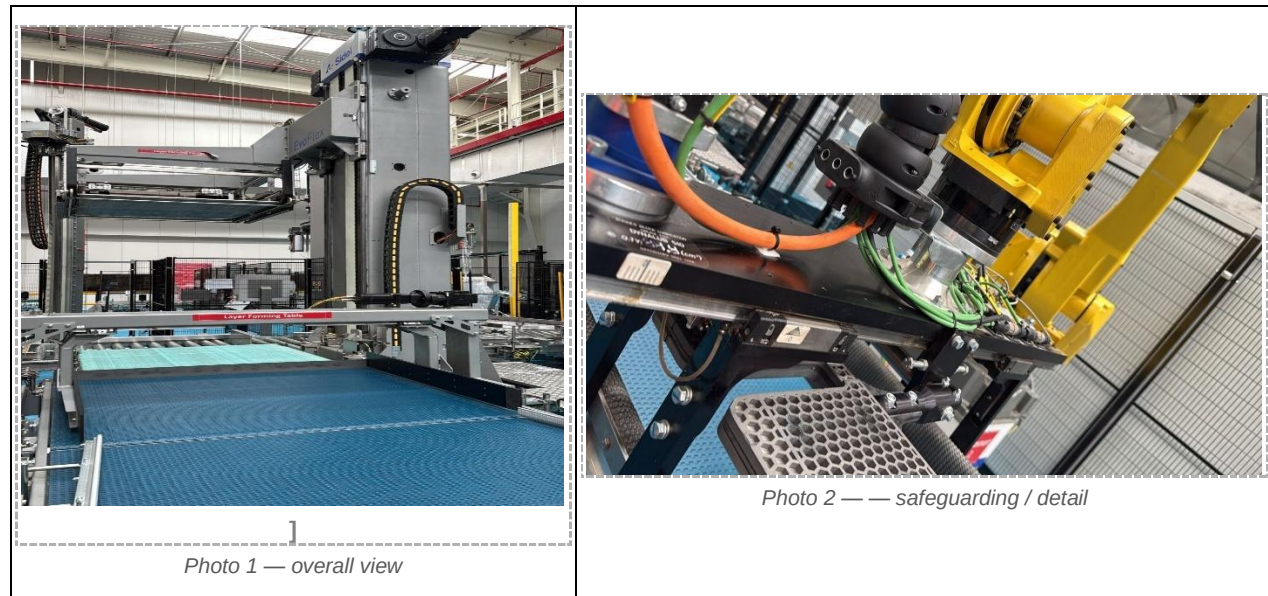


Figure HZ-07: Adjustable guide mechanism

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Guide adjuster handles in safe position	Existing
Device	T015	Mode selector for manual mode during adjustment	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO before guide adjuster maintenance	Critical
Admin	A004	Written changeover SOP specifying manual mode	Recommended

► ACTION REQUIRED

- [A013] LOTO before guide adjuster maintenance — *Critical*
- [A004] Written changeover SOP specifying manual mode — *Recommended*

Hazard Zone HZ-08: General Moving Parts – Awareness

Hazard Source:	Multiple simultaneously moving assemblies	Consequence:	Loss of Awareness / Ergonomic
Risk IDs:	R033-R035	Tasks:	T01,T11,T13
Risk Parameters:	S1 / F2 / P2 => PLr c		

Rotate operators on repetitive tasks. Assess ergonomic load of interlayer pad and pallet handling.

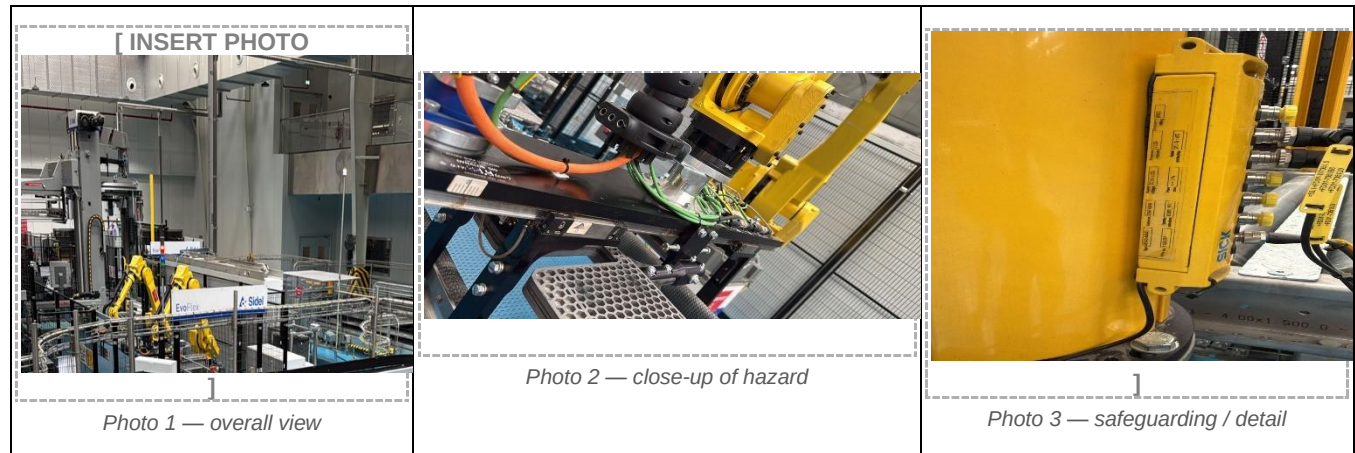


Figure HZ-08: Palletizer perimeter and machine area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Transparent guard panels where practical	Existing
Device	T001	Audible/visible warning before automatic start	Existing
Admin	A001	Stack lights/beacons during operation	Existing
Admin	A002	Awareness signs at access points	Existing
Admin	A019	Mandatory PPE	Existing

C.9 KORE Compliance – Palletizer

KORE compliance status for the Palletizer. Items marked Action Req. require client follow-up.

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	Full ISO 12100 assessment completed; 35+ hazard-task combinations.	Compliant	Sec. C.7
Machine Guarding	Perimeter guards, interlocked doors, light curtains adequate.	Compliant	C.7
LOTO	Electrical, pneumatic, and mechanical energy isolation procedures to be issued.	Action Req.	HZ-04, HZ-05
Electrical Safety	Main isolator, E-stops, dual earthing PE in place.	Compliant	HZ-04, HZ-06
Emergency Stop	Coverage adequate at operator stations.	Compliant	C.7
Training & Competency	Changeover SOPs to be issued and trained.	Action Req.	C.4, C.5
Inspection & PM	Periodic inspection programme to be set up.	Action Req.	C.7
Contractor & Visitor Safety	Visitor exposure assessed; signage in place.	Compliant	HZ-01
PPE	Safety footwear and hand protection mandatory.	Compliant	C.7
Management of Change	Modifications require re-assessment.	Advisory	-

C-Annex. Stretch Wrapper (Post-Palletizer)

This section covers the Robopac stretch wrapper installed downstream of the Palletizer. Wrapped pallets are dispatched to storage after this stage. Hazards are limited but contain one critical gap — no trapped-key system is presently in place on the machine.

TOP 3 PRIORITIES

- 1. Trapped-key system not installed** — Site finding: a standard machine key is currently used on the control box (not a trapped-key interlock). Replace with trapped-key per ISO 14119 (Type 4/5). Critical — sequential lockout cannot be enforced as-is.
- 2. Rotating wrapping arm — no-go zone during operation** — The wrapping arm rotates around stationary pallets at significant speed. Maintain interlocked perimeter guarding. No access during automatic operation.
- 3. Film carriage entanglement at low level** — Film carriage travels vertically and can trap fingers or hands near the rollers. Apply LOTO before any film roll changeover or jam clearing.

C-Annex.1 Machine Identification

Machine Name	Stretch Wrapper
Manufacturer	Robopac (Aetna Group SpA, Italy)
Machine Family	Stretch Wrapper
Serial Number	T123020066
Rated Voltage	400 Vac
Frequency / Phases	50 Hz / 3PH / PE
Absorption	40.6 A
Total Power	21.0 kW
CE Marking	CE Marked
Made In	Italy



Figure C-Annex.1: Robopac stretch wrapper nameplate

C-Annex.2 Overview

Vertical-mast Robopac stretch wrapper. The wrapping arm carries the film carriage around the stationary pallet to wrap the load. Pallets arrive from the Palletizer outfeed conveyor, are wrapped, and then dispatched.

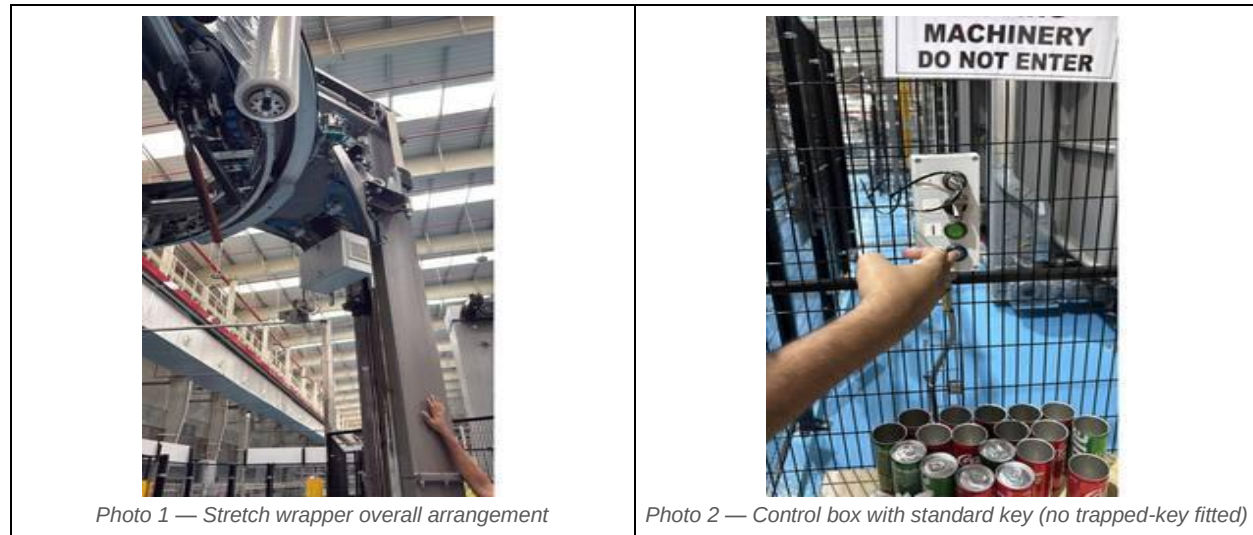


Figure C-Annex.2: Stretch wrapper — overall view and current key arrangement

C-Annex.3 Risk Reduction Measures

Category	Ref.	Measure	Status
Guard	G003	Perimeter guard with interlocked access door	Existing
Device	T011	E-Stop push button per ISO 13850	Existing
Device	T008	Local isolator with lockable OFF position	Existing
Device	T014	Interlock with panic release per ISO 14119, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d — currently NOT FITTED. Standard machine key in use on the control box. Replace with trapped-key system.	Recommended addition
Admin	A013	Machine-specific LOTO procedure for film roll changeover and maintenance	Recommended corrections
Admin	A002	Warning signs at access point (MACHINERY DO NOT ENTER sign present)	Existing
Admin	A019	Mandatory PPE — safety footwear, hand protection	Existing

► ACTION REQUIRED

- [T022] Replace existing standard key with trapped-key interlock system per ISO 14119 (Type 4/5) — *Critical*
- [A013] Issue and train machine-specific LOTO procedure for film roll changeover and maintenance — *Recommended corrections*

D. Filler

High-speed rotary filler. Fills product into empty cans. Seven hazard zones; the carousel and starwheel zones are the main concerns.

TOP 3 PRIORITIES

- 1. Machine-specific LOTO procedure** — Cover electrical isolation, rotational standstill verification on the carousel, and starwheel safe access.
- 2. Verify dual PE conductors on all motor bodies** — Add body earthing where missing. Test continuity. Document each motor.
- 3. Issue task SOPs for jam clearing and changeover** — Carousel jam clearing, starwheel guide adjustment, format changeover. Train operators.

D.1 Machine Identification

Machine Name	Filler
Manufacturer	SIDEL
Machine Family	FILLER
Type / Model	EVOFILL
Serial ID	SF330X340147
Year of Manufacture	2026
Rated Voltage	400 V
Full Load Current	50 A
Control Voltage	24 VDC
CE Marking	Non CE Marked

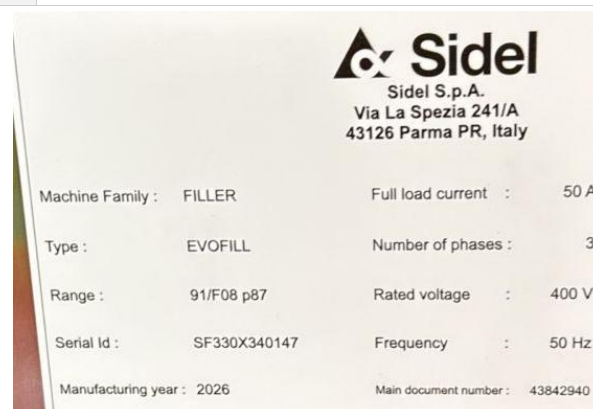


Figure D.1: Filler nameplate

D.2 Action Summary

Consolidated list of actions identified across all hazard zones for the Filler. Detail and supporting context follow in subsequent sections.

Zone	Ref.	Action	Status
HZ-01	A013	LOTO procedures for maintenance and cleaning	Recommended corrections
HZ-02	A013	LOTO mandatory for all work within carousel envelope	Recommended correction
HZ-03	A013	LOTO before jam clearing or guide adjustment	Critical
HZ-04	A013	Machine-specific LOTO SOP	Critical
HZ-04	A004	Live work by qualified electricians only	Recommended
HZ-05	A013	LOTO with standstill verification	Critical
HZ-06	A013	LOTO before any electrical work	Critical
HZ-06	A006	Annual earth continuity / insulation resistance testing	Critical

D.3 Overview & Process

Rotary filler with infeed starwheel, filling carousel with multiple filling valves, and discharge starwheel. Sits between the rinser and the seamer.

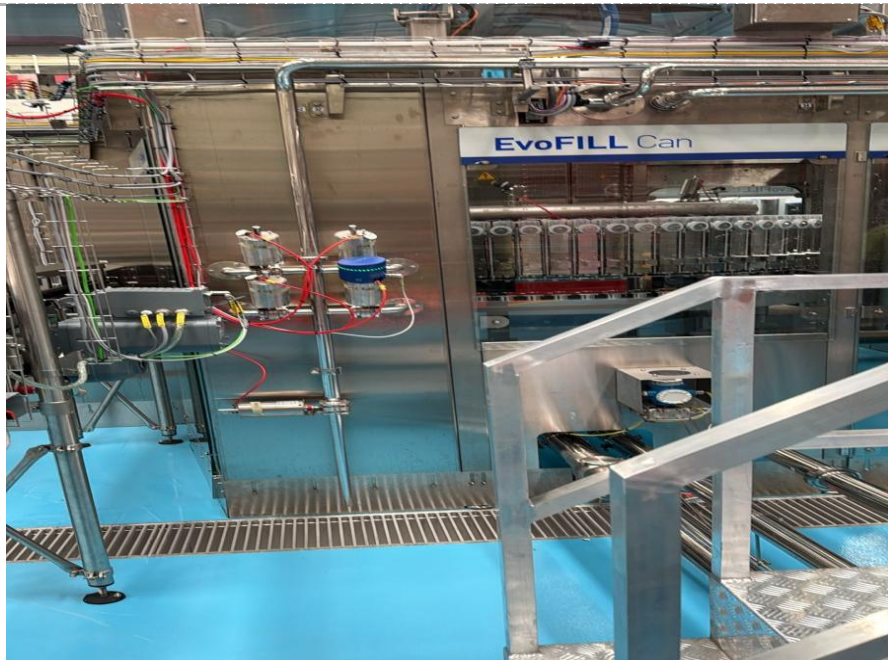


Figure D.2: Filler – machine overview

Empty cans arrive on the infeed conveyor and are picked up by the infeed starwheel. They transfer into the carousel, get positioned under a filling valve, the valve fills the can, and the discharge starwheel takes them out onto the discharge conveyor.

D.4 Operating Modes

Mode	Description
Automatic	Normal production. PLC controls all motion. Operator only monitors and clears jams from designated points.
Manual / Jog	Reduced-speed jog under operator control. Used for setup, fault-finding, and changeover.

D.5 Limits of the Machine

Intended Use

Filling beverage product into empty cans during normal production.

Foreseeable Misuse

Entering the carousel zone while in automatic mode without LOTO. Disabling interlocks or light curtains. Skipping energy isolation during maintenance. Bypassing guard door interlocks. Climbing on machine structures. HMI override of safety parameters.

Known Faults

Sensor faults causing positioning errors. Mechanical blockage from misaligned cans or foreign objects. Control system faults causing unintended movement.

D.6 Tasks

Tasks observed during the site visit. These cover both routine production and maintenance activity.

No.	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor adjustment / fault-finding	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans	Operator	Stopped	As needed
T06	Manual jog of carousel / starwheel	Operator	Manual	As needed
T07	Inspection of carousel	Operator	Auto	Continuous
T08	HMI parameter changes	Maintenance	Manual	Periodic
T09	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T10	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T11	Setting / changeover	Maintenance	Manual	Periodic

D.7 Risk Summary

Zone	Hazard Source	Consequence	Risk IDs	Tasks	S/F/P	PLr	Achieved
HZ-01	Infeed & Discharge Conveyors	Crushing	R001-R004	T01-T04	S2/F2/P1	D	Compliant
HZ-02	Filling Carousel	Crushing	R005-R009	T02,T05-T07	S2/F2/P2	E	Compliant
HZ-03	Starwheel / Can Transfer	Crushing	R010-R014	T02,T05,T07	S2/F1/P1	C	Compliant
HZ-04	Control Panel / Live Terminals	Electrocution	R015-R019	T02,T05,T06	S2/F1/P2	NA	Compliant
HZ-05	Drive Motors & Gearboxes	Entanglement	R020-R024	T02,T05,T11	S2/F1/P1	C	Compliant
HZ-06	Electrical Earthing	Electrocution	R025-R030	T02,T05,T06	S2/F2/P1	NA	Compliant
HZ-07	General Moving Parts	Loss of Awareness	R031-R033	T01,T11	S1/F2/P2	NA	Compliant

D.8 Hazard Zones

Hazard Zone HZ-01: Infeed & Discharge Conveyors

Hazard Source:	Roller / chain conveyors	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R004	Tasks:	T01,T02,T03,T04
Risk Parameters:	S2 / F2 / P1 => PLr d		

Maintain interlocked guarding at chain drive nip points. Apply LOTO before jam-clearing inside the safeguarded space.

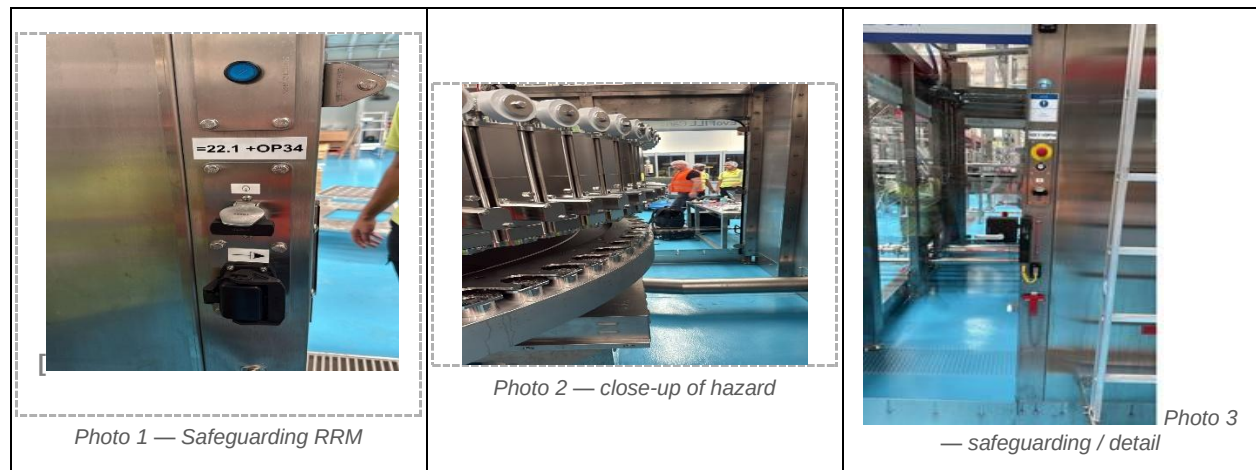


Figure HZ-01: Filler infeed conveyor

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges; non-slip surfaces in operator zones	Existing
Guard	G003	Perimeter guard with interlocks and reset	Existing
Guard	G004	Movable guard (hinged access door) per ISO 14120	Existing
Device	T011	E-Stop push buttons, PLr d	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T005	Reset button (blue) with indicator	Existing
Admin	A013	LOTO procedures for maintenance and cleaning	Recommended corrections
Admin	A002	Awareness signs at access points	Existing

► ACTION REQUIRED

- [A013] LOTO procedures for maintenance and cleaning — Recommended corrections

Hazard Zone HZ-02: Filling Carousel

Hazard Source:	Filling carousel rotating assembly	Consequence:	Crushing / Entanglement
Risk IDs:	R005-R009	Tasks:	T02, T05, T06, T07
Risk Parameters:	S2 / F2 / P2 => PLr e		

Big rotating structure with multiple filling valves and starwheels. Mass and inertia mean once you're caught there's no avoiding contact. Maintain PLr d interlocks and PLr e light curtains. Test functional integrity quarterly.

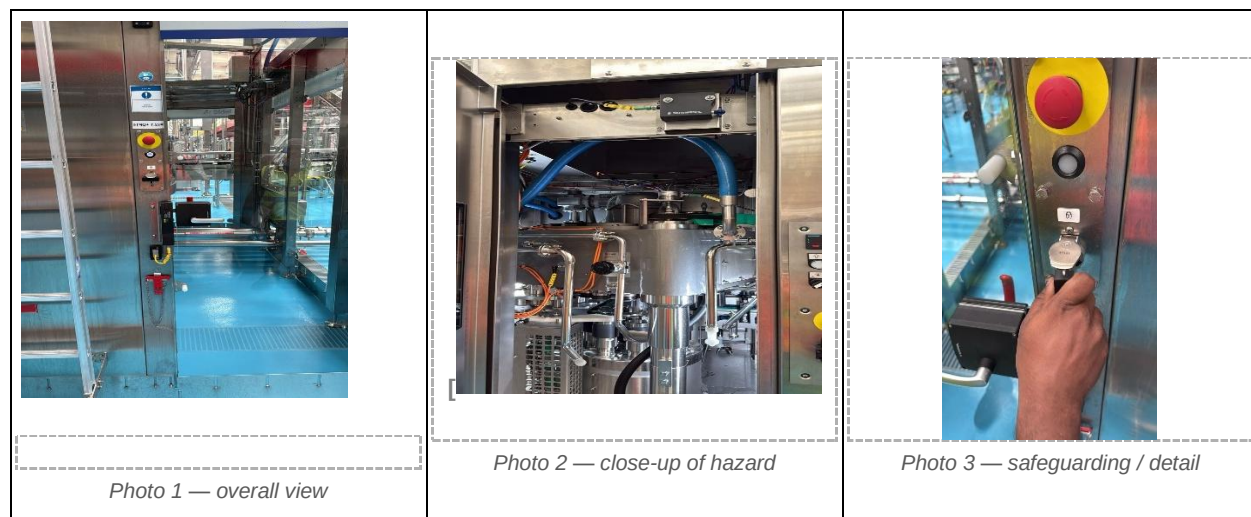


Figure HZ-02: Filling carousel area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical stops and physical barriers	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Guard	G004	Movable guard (hinged access door)	Existing
Device	T010	Safety light curtain per ISO 13855, PLr e	Existing
Device	T014	Interlock with panic release on all access doors, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing
Device	T005	Reset button inside safeguarded space, PLr e	Existing
Device	T011	E-Stop push buttons	Existing
Device	T015	Mode selector for manual mode, PLr c	Existing
Admin	A013	LOTO mandatory for all work within carousel envelope	Recommended correction
Admin	A007	Measures to avoid unauthorised access	Existing

► ACTION REQUIRED

- [A013] LOTO mandatory for all work within carousel envelope — Recommended correction

Hazard Zone HZ-03: Starwheel & Can Transfer

Hazard Source:	Starwheels, guides & can transfer	Consequence:	Crushing / Impact
Risk IDs:	R010-R014	Tasks:	T02,T05,T07,T10
Risk Parameters:	S2 / F1 / P1 => PLr c		

Apply LOTO before clearing jams or adjusting guides at starwheel transfer points. Mandatory hand protection.

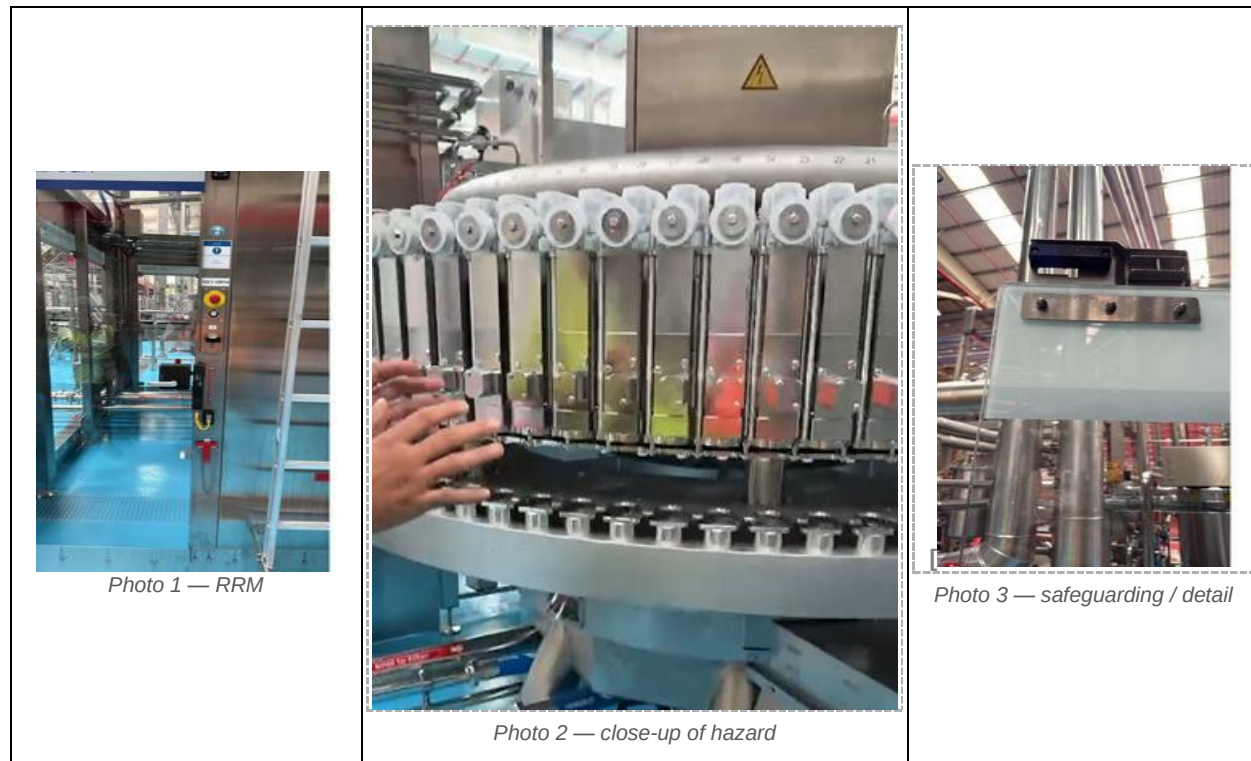


Figure HZ-03: Starwheel and can transfer

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Safe geometry; minimum gap per ISO 13854	Existing
Guard	G003	Perimeter guard preventing access during automatic mode	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO before jam clearing or guide adjustment	Critical
Admin	A019	Mandatory hand and foot protection	Existing

► ACTION REQUIRED

- [A013] LOTO before jam clearing or guide adjustment — Critical

Hazard Zone HZ-04: Control Panel & Live Terminals

Hazard Source:	Control panel live terminals	Consequence:	Electrocution
Risk IDs:	R015-R019	Tasks:	T02,T05,T06,T09,T11
Risk Parameters:	S2 / F1 / P2 => PLr d		

Open panels only by qualified electricians. Identify permanently live circuits in the LOTO procedure. Verify warning labels in place.



Figure HZ-04: Filler main electrical panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	IP-rated enclosures	Existing
Guard	G001	Fixed guard on control panel	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T021	Dual earthing PE protection	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	Machine-specific LOTO SOP	Critical
Admin	A004	Live work by qualified electricians only	Recommended

► ACTION REQUIRED

- [A013] Machine-specific LOTO SOP — *Critical*
- [A004] Live work by qualified electricians only — *Recommended*

Hazard Zone HZ-05: Drive Motors & Gearboxes

Hazard Source:	Drive motors, gearboxes & belts	Consequence:	Entanglement / Impact
Risk IDs:	R020-R024	Tasks:	T02,T05,T11,T13
Risk Parameters:	S2 / F1 / P1 => PLr c		

Drives for the carousel and conveyors. Belts and chains must stay enclosed. Verify standstill before any maintenance access — rotational inertia continues after electrical isolation. *Figure HZ-05: Drive motor and timing belt assembly*

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Fixed guarding on belt, chain and shaft drives	Existing
Guard	G001	Fixed guard per ISO 14120 on all drive assemblies	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO with standstill verification	Critical
Admin	A006	Periodic inspection of drive guards and fixings	Existing

► ACTION REQUIRED

- [A013] LOTO with standstill verification — Critical

Hazard Zone HZ-06: Electrical Earthing & Over-Voltage

Hazard Source:	Electrical earthing / over-voltage	Consequence:	Electrocution / Shock
Risk IDs:	R025-R030	Tasks:	T02,T05,T06,T10,T11,T13
Risk Parameters:	S2 / F2 / P1 => PLr d		

Verify two independent PE conductors per IEC 60204-1. Schedule annual continuity testing. Confirm surge protection in the panel is functional.

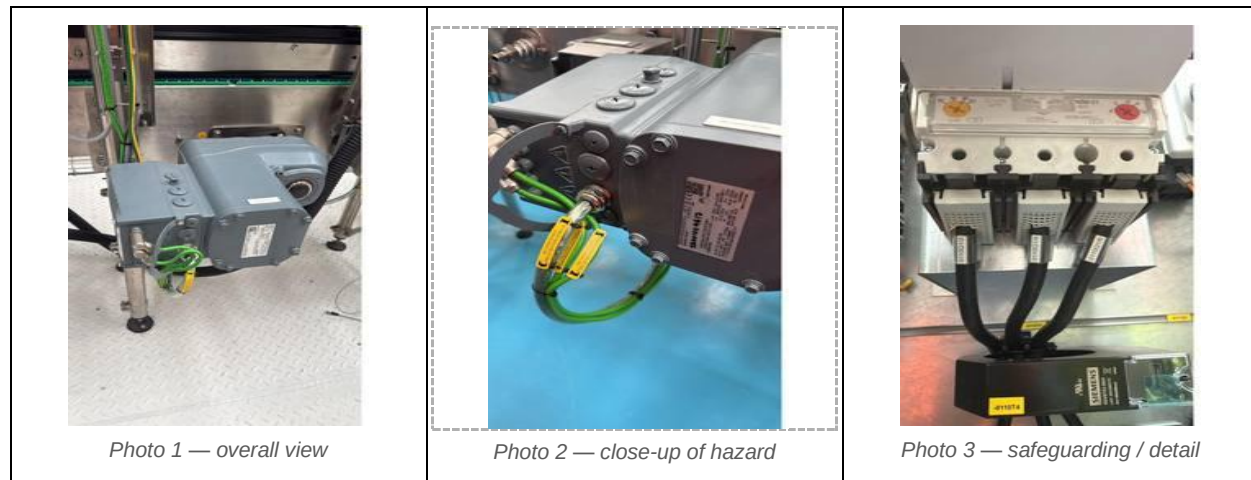


Figure HZ-06: Electrical earthing arrangement

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Equipotential bonding of all metalwork	Existing at motors – body earthing to be added
Device	T021	Dual earthing PE protection	Existing
Device	T019	Over-voltage protection device	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Admin	A013	LOTO before any electrical work	Critical
Admin	A006	Annual earth continuity / insulation resistance testing	Critical

► ACTION REQUIRED

- [A013] LOTO before any electrical work — Critical
- [A006] Annual earth continuity / insulation resistance testing — Critical

Hazard Zone HZ-07: General Moving Parts – Awareness

Hazard Source:	Multiple simultaneously moving parts	Consequence:	Loss of Awareness / Ergonomic
Risk IDs:	R031-R033	Tasks:	T01,T11,T13
Risk Parameters:	S1 / F2 / P2 => PLr c		

Carousel, starwheels, conveyors all moving at once. Habituation and ergonomic strain over time. *Figure HZ-07: General machine area*

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Transparent guard panels where practical	Existing
Device	T001	Audible/visible warning before automatic start	Existing
Admin	A001	Stack lights/beacons during operation	Existing
Admin	A002	Awareness signs at access points	Existing
Admin	A019	Mandatory PPE	Existing

D.9 KORE Compliance – Filler

KORE compliance status for the Filler. Items marked Action Req. require client follow-up.

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	Full ISO 12100 assessment; 33+ hazard-task combinations.	Compliant	Sec. D.7
Machine Guarding	Perimeter guards, interlocked doors, light curtains adequate.	Compliant	D.7
LOTO	Electrical and mechanical energy isolation. Machine-specific SOP needed.	Action Req.	HZ-04
Electrical Safety	Main isolator, E-stops, dual earthing PE in place.	Compliant	HZ-04, HZ-06
Emergency Stop	Coverage adequate.	Compliant	D.7
Training & Competency	Task SOPs to be issued.	Action Req.	D.4, D.5
Inspection & PM	Periodic inspection programme to be set up.	Action Req.	D.7
Contractor & Visitor Safety	Visitor exposure assessed.	Compliant	HZ-01
PPE	Safety footwear, ear and hand protection mandatory.	Compliant	D.7
Management of Change	Modifications require re-assessment.	Advisory	-

E. Can Seamer

Double-seams lids onto filled cans. Eight hazard zones; the seaming turret and seaming heads carry the highest residual risk.

TOP 3 PRIORITIES	
1. Machine-specific LOTO including seaming head changeover	— Critical for seaming roll and chuck adjustment. Cover electrical and mechanical isolation. Train maintenance team.
2. Issue written SOP for seaming roll/chuck adjustment	— Step-by-step procedure with safe access sequence. Mandatory before any chuck change.
3. Annual earth continuity test programme	— Schedule and document annual continuity and insulation resistance testing for all panels and motors.

E.1 Machine Identification

Machine Name	Can Seamer
Manufacturer	[refer to nameplate photo]
Machine Family	CAN SEAMER
Type / Model	NA
Serial ID	NA
Year of Manufacture	2026
Rated Voltage	400 V
Full Load Current	50 A
Control Voltage	24 VDC
CE Marking	Non CE Marked

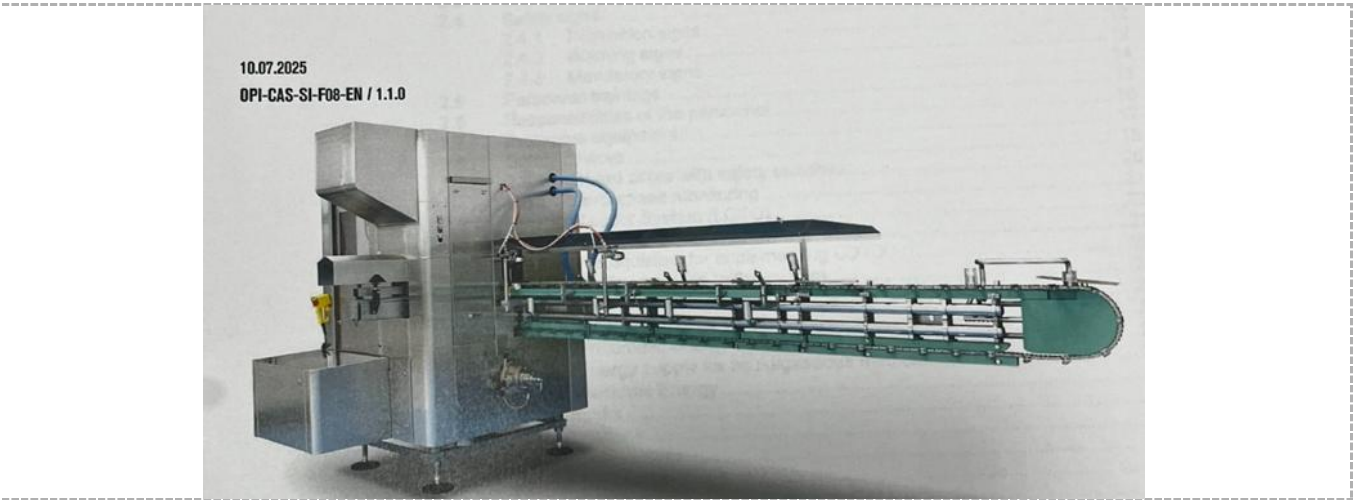


Figure E.1: Can Seamer nameplate

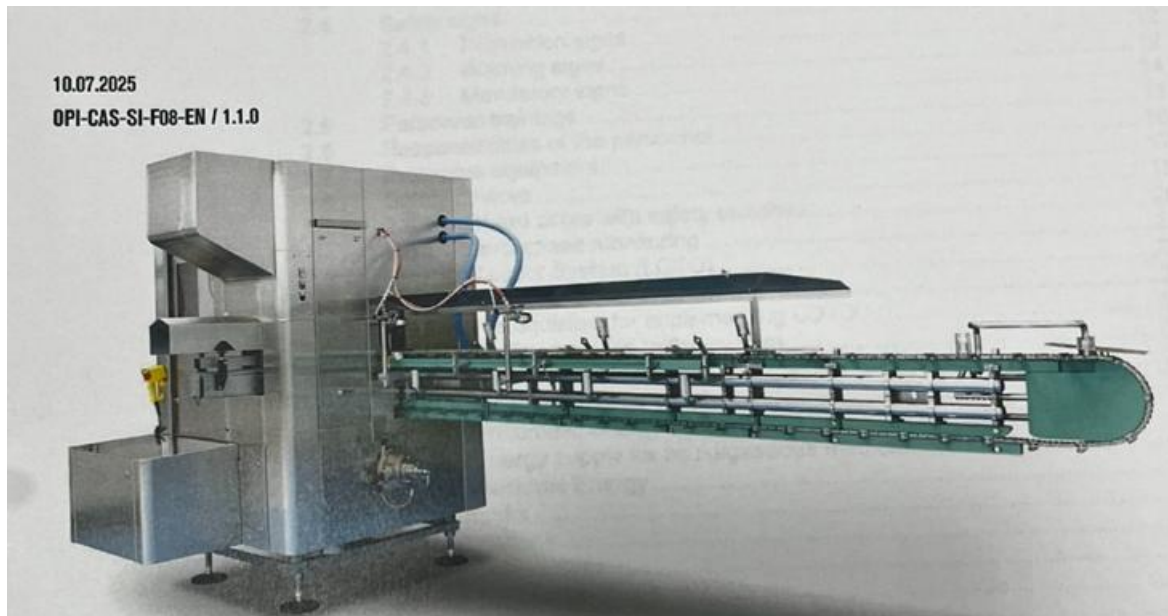
E.2 Action Summary

Consolidated list of actions identified across all hazard zones for the Can Seamer. Detail and supporting context follow in subsequent sections.

Zone	Ref.	Action	Status
HZ-01	A013	LOTO procedures for maintenance and cleaning	Recommended corrections
HZ-02	A013	LOTO mandatory for all work within turret envelope	Recommended correction
HZ-03	A013	LOTO before seaming head maintenance / changeover	Critical
HZ-03	A004	Written SOP for seaming roll and chuck adjustment	Recommended
HZ-04	A013	LOTO before jam clearing in lid feed zone	Critical
HZ-05	A013	Machine-specific LOTO SOP	Critical
HZ-05	A004	Live work by qualified electricians only	Recommended
HZ-06	A013	LOTO with standstill verification	Critical
HZ-07	A013	LOTO before any electrical work	Critical
HZ-07	A006	Annual earth continuity / insulation resistance testing	Critical

E.3 Overview & Process

Rotary double-seamer with infeed starwheel, seaming turret with multiple seaming heads, lid feed system, and discharge starwheel. Sits immediately downstream of the filler.



Seamer – machine overview

Figure E.2: Can

Filled cans arrive from the filler. Lids feed from the hopper via the feed track and are placed on each can. Cans transfer into the seaming turret where each head performs the double-seam. Sealed cans discharge to the outfeed conveyor.

E.4 Operating Modes

Mode	Description
Automatic	Normal production. PLC controls all motion. Operator only monitors and clears jams from designated points.
Manual / Jog	Reduced-speed jog under operator control. Used for setup, fault-finding, and changeover.

E.5 Limits of the Machine

Intended Use

Applying and double-seaming lids onto filled cans during normal production.

Foreseeable Misuse

Entering the seaming turret zone while in automatic mode without LOTO. Disabling interlocks. Skipping LOTO during seaming head maintenance. Bypassing guard door interlocks. HMI override of safety parameters.

Known Faults

Lid jam in the feed track. Seaming roll wear giving defective seams. Misaligned cans in the infeed jamming the turret. Control system faults.

E.6 Tasks

Tasks observed during the site visit. These cover both routine production and maintenance activity.

No.	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor adjustment / fault-finding	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans / lids	Operator	Stopped	As needed
T06	Manual jog of seaming turret	Operator	Manual	As needed
T07	Inspection of seaming turret	Operator	Auto	Continuous
T08	Lid replenishment	Operator	Auto	Continuous
T09	HMI parameter changes	Maintenance	Manual	Periodic
T10	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T11	Seaming roll / chuck changeover	Maintenance	LOTO	Periodic
T12	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T13	Setting / changeover	Maintenance	Manual	Periodic

E.7 Risk Summary

Zone	Hazard Source	Consequence	Risk IDs	Tasks	S/F/P	PLr	Achieved
HZ-01	Infeed & Discharge Conveyors	Crushing	R001-R004	T01-T04	S2/F2/P1	D	Compliant
HZ-02	Seaming Turret	Crushing	R005-R009	T02,T05-T07	S2/F2/P2	E	Compliant
HZ-03	Seaming Heads & Chuck	Crushing	R010-R015	T02,T05,T07,T10	S2/F2/P2	E	Compliant
HZ-04	Lid Feed & Hopper	Crushing/Shear	R016-R019	T02,T05,T07	S2/F1/P1	C	Compliant
HZ-05	Control Panel / Live Terminals	Electrocution	R020-R024	T02,T05,T06	S2/F1/P2	NA	Compliant
HZ-06	Drive Motors & Gearboxes	Entanglement	R025-R029	T02,T05,T11	S2/F1/P1	C	Compliant
HZ-07	Electrical Earthing	Electrocution	R030-R035	T02,T05,T06	S2/F2/P1	NA	Compliant
HZ-08	General Moving Parts	Loss of Awareness	R036-R038	T01,T11	S1/F2/P2	NA	Compliant

E.8 Hazard Zones

Hazard Zone HZ-01: Infeed & Discharge Conveyors

Hazard Source:	Roller / chain conveyors	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R004	Tasks:	T01,T02,T03,T04
Risk Parameters:	S2 / F2 / P1 => PLr d		

Apply standard conveyor controls — interlocked nip-point guarding, LOTO for any access.

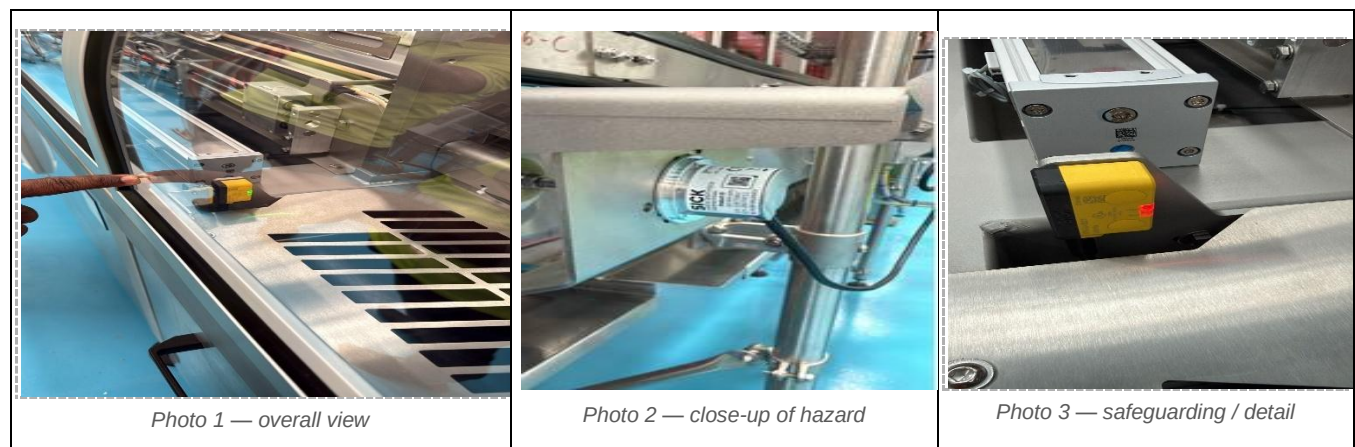


Figure HZ-01: Seamer infeed conveyor

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges; non-slip surfaces in operator zones	Existing
Guard	G003	Perimeter guard with interlocks and reset	Existing
Guard	G004	Movable guard (hinged access door)	Existing
Device	T011	E-Stop push buttons, PLr d	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T005	Reset button (blue) with indicator	Existing
Admin	A013	LOTO procedures for maintenance and cleaning	Recommended corrections
Admin	A002	Awareness signs at access points	Existing

► ACTION REQUIRED

- [A013] LOTO procedures for maintenance and cleaning — Recommended corrections

Hazard Zone HZ-02: Seaming Turret

Hazard Source:	Seaming turret rotating assembly	Consequence:	Crushing / Impact
Risk IDs:	R005-R009	Tasks:	T02, T05, T06, T07
Risk Parameters:	S2 / F2 / P2 => PLr e		

High-speed rotating turret holding the seaming heads. Maintain interlocked perimeter at PLr d. Once contact occurs, avoidance is not possible — access during operation must be physically prevented.



Photo 1 — overall view

Figure HZ-02: Seaming turret

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Mechanical stops and physical barriers	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Guard	G004	Movable guard (hinged access door)	Existing
Device	T010	Safety light curtain, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T005	Reset button inside safeguarded space, PLr e	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO mandatory for all work within turret envelope	Recommended correction

► ACTION REQUIRED

- [A013] LOTO mandatory for all work within turret envelope — Recommended correction

Hazard Zone HZ-03: Seaming Heads & Chuck Assembly

Hazard Source:	Seaming heads, rolls & chuck assembly	Consequence:	Crushing / Entanglement
Risk IDs:	R010-R015	Tasks:	T02, T05, T07, T10, T11
Risk Parameters:	S2 / F2 / P2 => PLr e		

First and second operation seaming rolls plus chucks. Concentrated force during double-seaming. Apply full LOTO before setup, changeover, or maintenance. No exceptions.



Photo 1 — overall view

Figure HZ-03: Seaming head area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Safe geometry; minimum gaps per ISO 13854	Existing
Guard	G003	Perimeter guard preventing access during automatic mode	Existing
Guard	G004	Movable guard (hinged)	Existing
Device	T010	Safety light curtain, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing
Device	T015	Mode selector for manual mode	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO before seaming head maintenance / changeover	Critical
Admin	A004	Written SOP for seaming roll and chuck adjustment	Recommended

► ACTION REQUIRED

- [A013] LOTO before seaming head maintenance / changeover — Critical
- [A004] Written SOP for seaming roll and chuck adjustment — Recommended

Hazard Zone HZ-04: Control Panel & Live Terminals

Hazard Source:	Control panel live terminals	Consequence:	Electrocution
Risk IDs:	R020-R024	Tasks:	T02, T05, T06, T09, T11
Risk Parameters:	S2 / F1 / P2 => PLr d		

400V terminals in the main panel.



Photo 1 — overall view



Photo 3 — safeguarding / detail

Figure HZ-05: Seamer main electrical panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	IP-rated enclosures	Existing
Guard	G001	Fixed guard on control panel	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T021	Dual earthing PE protection	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	Machine-specific LOTO SOP	Critical
Admin	A004	Live work by qualified electricians only	Recommended

► ACTION REQUIRED

- [A013] Machine-specific LOTO SOP — *Critical*
- [A004] Live work by qualified electricians only — *Recommended*

Hazard Zone HZ-05: Drive Motors & Gearboxes

Hazard Source:	Drive motors, gearboxes & belts	Consequence:	Entanglement / Impact
Risk IDs:	R025-R029	Tasks:	T02, T05, T11, T13
Risk Parameters:	S2 / F1 / P1 => PLr c		

Keep all belt and chain drives fully enclosed. Apply LOTO with standstill verification before drive maintenance.



Photo 1 — overall view

Figure HZ-06: Drive motor and timing belt

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Fixed guarding on belt, chain and shaft drives	Existing
Guard	G001	Fixed guard per ISO 14120 on drive assemblies	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO with standstill verification	Critical

► ACTION REQUIRED

- [A013] LOTO with standstill verification — Critical

Hazard Zone HZ-06: Electrical Earthing & Over-Voltage

Hazard Source:	Electrical earthing / over-voltage	Consequence:	Electrocution / Shock
Risk IDs:	R030-R035	Tasks:	T02,T05,T06,T10,T11,T13
Risk Parameters:	S2 / F2 / P1 => PLr d		

Verify two PE conductors per IEC 60204-1. Test surge protection. Schedule annual continuity testing.



Photo 1 — overall view

Figure HZ-07: Electrical earthing

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Equipotential bonding of all metalwork	Existing at motors – body earthing to be added
Device	T021	Dual earthing PE protection	Existing
Device	T019	Over-voltage protection device	Existing

Device	T008	Main isolator (lockable OFF)	Existing
Admin	A013	LOTO before any electrical work	Critical
Admin	A006	Annual earth continuity / insulation resistance testing	Critical

► ACTION REQUIRED

- [A013] LOTO before any electrical work — Critical
- [A006] Annual earth continuity / insulation resistance testing — Critical

Hazard Zone HZ-07: General Moving Parts – Awareness

Hazard Source:	Multiple simultaneously moving parts	Consequence:	Loss of Awareness / Ergonomic
Risk IDs:	R036-R038	Tasks:	T01,T11,T13
Risk Parameters:	S1 / F2 / P2 => PLr c		

Rotate operators on repetitive tasks. Schedule ergonomic assessment of jam clearing and manual handling positions.

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Transparent guard panels where practical	Existing
Device	T001	Audible/visible warning before automatic start	Existing
Admin	A001	Stack lights/beacons during operation	Existing
Admin	A002	Awareness signs at access points	Existing
Admin	A019	Mandatory PPE	Existing

E.9 KORE Compliance – Can Seamer

KORE compliance status for the Can Seamer. Items marked Action Req. require client follow-up.

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	Full ISO 12100 assessment; 38+ hazard-task combinations.	Compliant	Sec. E.7
Machine Guarding	Perimeter guards, interlocked doors, light curtains adequate.	Compliant	E.7
LOTO	Electrical and mechanical isolation including seaming head changeover.	Action Req.	HZ-03, HZ-05
Electrical Safety	Main isolator, E-stops, dual earthing PE in place.	Compliant	HZ-05, HZ-07
Emergency Stop	Coverage adequate.	Compliant	E.7
Training & Competency	Seaming changeover SOPs to be issued.	Action Req.	E.4, E.5
Inspection & PM	Periodic inspection programme to be set up.	Action Req.	E.7
Contractor & Visitor Safety	Visitor exposure assessed.	Compliant	HZ-01
PPE	Safety footwear, ear and hand protection mandatory.	Compliant	E.7
Management of Change	Modifications require re-assessment.	Advisory	-

F. OCME Packer

Film wrapper. Groups filled cans and wraps them in plastic film to form multipacks. Eight hazard zones; the film carriage and sealing bar are the highest residual risk.

TOP 3 PRIORITIES

- 1. Flag permanently live circuits in main panel** — Verify warning labels on all panels. Update LOTO procedure cross-reference. Critical — existing main isolator does not isolate all circuits.
- 2. Machine-specific LOTO with thermal cooldown** — Sealing bar requires pneumatic isolation and thermal cooldown wait before access. Write SOP and train maintenance.
- 3. Issue film-handling SOPs** — Film roll changeover, film threading, sealing bar maintenance. Mount laminated work cards at each operator station.

F.1 Machine Identification

Machine Name	OCME Packer
Manufacturer	OCME S.r.l., Parma, Italy
Machine Family	PACKER / FILM WRAPPER
Type / Model	Heating Tunnel Cabinet
Serial ID	2500010A01
Year of Manufacture	2026
Rated Voltage	400 V
Full Load Current	50 A
Control Voltage	24 VDC
CE Marking	Non CE Marked



Figure F.1: OCME Packer nameplate

F.2 Action Summary

Consolidated list of actions identified across all hazard zones for the OCME Packer. Detail and supporting context follow in subsequent sections.

Zone	Ref.	Action	Status
HZ-01	A013	LOTO procedures for maintenance and cleaning	Recommended corrections
HZ-02	A013	LOTO before all work within carriage sweep envelope	Recommended correction
HZ-03	A013	LOTO with pneumatic isolation and thermal cooldown	Critical
HZ-04	A013	LOTO with standstill verification	Critical
HZ-05	A013	LOTO with pneumatic isolation and bleed-down	Critical
HZ-06	A013	Machine-specific LOTO – flag permanently live circuits	Critical
HZ-06	A004	Live work by qualified electricians only	Recommended
HZ-07	A013	LOTO before any electrical work	Critical
HZ-07	A006	Annual earth continuity / insulation resistance testing	Critical

F.3 Overview & Process

Film wrapper with infeed conveyor (light curtain protected), overhead circular film delivery carriage with film rolls, pneumatically actuated heated sealing/cutting bar, drive motors, and discharge conveyor.



Figure F.2: OCME Packer – machine overview

Filled can groups arrive from upstream, get metered into the packer, and the overhead carriage rotates dispensing film around them. The heated sealing bar descends to seal and cut the film at the tail of each pack. Wrapped packs discharge onto the outfeed conveyor.

F.4 Operating Modes

Mode	Description
Automatic	Normal production. PLC controls all motion. Operator only monitors and clears jams from designated points.
Manual / Jog	Reduced-speed jog under operator control. Used for setup, fault-finding, and changeover.

F.5 Limits of the Machine

Intended Use

Grouping and wrapping filled cans in plastic film to form multipacks during normal production.

Foreseeable Misuse

Entering the film carriage zone during automatic operation without LOTO. Disabling light curtains or door interlocks. Skipping LOTO including pneumatic isolation and thermal cooldown before sealing bar access. Climbing on the structure. Bypassing safety interlocks.

Known Faults

Film breakage or jam during carriage rotation. Sealing bar fails to close or open. Pneumatic actuator faults. Misaligned cans on the infeed jamming the conveyor. Control system errors.

F.6 Tasks

Tasks observed during the site visit. These cover both routine production and maintenance activity.

No.	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor / device adjustment	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans / film	Operator	Stopped	As needed
T06	Manual jog of carriage / conveyors	Operator	Manual	As needed
T07	Film roll replenishment / threading	Operator	Manual / Stopped	As needed
T08	Sealing bar adjustment / replacement	Maintenance	Manual / LOTO	Periodic
T09	HMI parameter changes	Maintenance	Manual	Periodic
T10	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T11	Drive belt / gearbox maintenance	Maintenance	LOTO	Scheduled
T12	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T13	Format / changeover	Maintenance	Manual	Periodic

F.7 Risk Summary

Zone	Hazard Source	Consequence	Risk IDs	Tasks	S/F/P	PLr	Achieved
HZ-01	Infeed & Discharge Conveyors	Crushing	R001-R004	T01-T04	S2/F2/P1	D	Compliant
HZ-02	Film Carriage (Overhead)	Crushing	R005-R009	T02,T05-T07	S2/F2/P2	E	Compliant
HZ-03	Film Sealing / Cutting Bar	Crushing/Burn	R010-R015	T02,T05,T07,T10	S2/F2/P2	E	Compliant
HZ-04	Drive Motors & Belts	Entanglement	R016-R020	T02,T05,T11	S2/F1/P1	C	Compliant
HZ-05	Pneumatic System	Impact/Crushing	R021-R025	T02,T05,T10	S2/F1/P1	C	Compliant
HZ-06	Control Panel / Live Terminals	Electrocution	R026-R030	T02,T05,T06	S2/F1/P2	NA	Compliant
HZ-07	Electrical Earthing	Electrocution	R031-R036	T02,T05,T06	S2/F2/P1	NA	Compliant
HZ-08	General Moving Parts	Loss of Awareness	R037-R039	T01,T11	S1/F2/P2	NA	Compliant

F.8 Hazard Zones

Hazard Zone HZ-01: Infeed & Discharge Conveyors

Hazard Source:	Roller / chain conveyors with safety light curtain columns	Consequence:	Crushing / Entanglement
Risk IDs:	R001-R004	Tasks:	T01,T02,T03,T04
Risk Parameters:	S2 / F2 / P1 => PLr d		

Maintain standard conveyor risk controls.



Figure HZ-01: Packer infeed conveyor

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Rounded edges; non-slip surfaces in operator zones	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Device	T010	Safety light curtains at infeed and discharge, PLr e	Existing
Device	T009	Cross-beam muting per IEC 62046	Existing
Device	T011	E-Stop push buttons, PLr d	Existing
Device	T014	Interlocked access doors with panic release, PLr d	Existing
Device	T005	Reset button (blue) with indicator	Existing
Admin	A013	LOTO procedures for maintenance and cleaning	Recommended corrections

► ACTION REQUIRED

- [A013] LOTO procedures for maintenance and cleaning — *Recommended corrections*

Hazard Zone HZ-02: Film Carriage (Overhead Rotating Assembly)

Hazard Source:	Overhead circular film delivery carriage	Consequence:	Crushing / Entanglement
Risk IDs:	R005-R009	Tasks:	T02, T05, T06, T07
Risk Parameters:	S2 / F2 / P2 => PLr e		

Treat the rotating carriage zone as no-go during automatic operation. Issue a written film roll changeover SOP — do not handle ad-hoc. Use elevated platform with appropriate fall protection.



Figure HZ-02: Overhead film delivery carriage

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Physical barriers enclosing carriage sweep zone	Existing
Guard	G003	Perimeter guard with interlocked access doors	Existing
Guard	G004	Movable guard (hinged access door)	Existing
Device	T010	Safety light curtain, PLr e	Existing
Device	T014	Interlocked access doors with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing
Device	T005	Reset button inside safeguarded space, PLr e	Existing
Device	T011	E-Stop push buttons	Existing
Device	T015	Mode selector key switch for manual mode, PLr c	Existing
Admin	A013	LOTO before all work within carriage sweep envelope	Recommended correction

► ACTION REQUIRED

- [A013] LOTO before all work within carriage sweep envelope — Recommended correction

Hazard Zone HZ-03: Film Sealing / Cutting Bar

Hazard Source:	Heated sealing bar and pneumatic cutting mechanism	Consequence:	Crushing / Shear / Burn
Risk IDs:	R010-R015	Tasks:	T02,T05,T07,T10,T11
Risk Parameters:	S2 / F2 / P2 => PLr e		

Heated sealing bar presents simultaneous crush, shear, and burn hazards. Apply LOTO with pneumatic isolation and thermal cooldown before any access. Heat-resistant gloves mandatory.

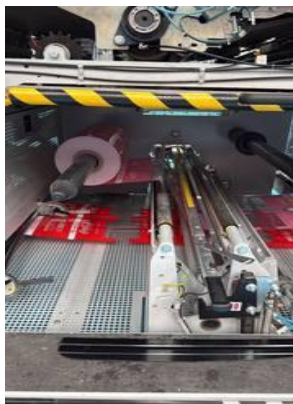


Photo 1 — overall view

Figure HZ-03: Film sealing/cutting bar mechanism

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Minimum gap per ISO 13854 on sealing bar guides	Existing
Guard	G003	Fixed and interlocked guards	Existing
Guard	G004	Hinged access door with interlock and panic release	Existing
Device	T010	Safety light curtain at sealing zone, PLr e	Existing
Device	T014	Interlock with panic release, PLr d	Existing
Device	T022	Trapped-key interlock per ISO 14119 (Type 4/5), PLr d	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO with pneumatic isolation and thermal cooldown	Critical
Admin	A019	Heat-resistant gloves when working near sealing bar	Existing

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation and thermal cooldown — Critical

Hazard Zone HZ-04: Drive Motors, Gearboxes & Belt Drives

Hazard Source:	Drive motors, gearboxes, timing belt drives	Consequence:	Entanglement / Impact
Risk IDs:	R016-R020	Tasks:	T02,T05,T11,T13
Risk Parameters:	S2 / F1 / P1 => PLr c		

Keep all belt and chain drives fully enclosed. Verify guard fixings require tools to remove.

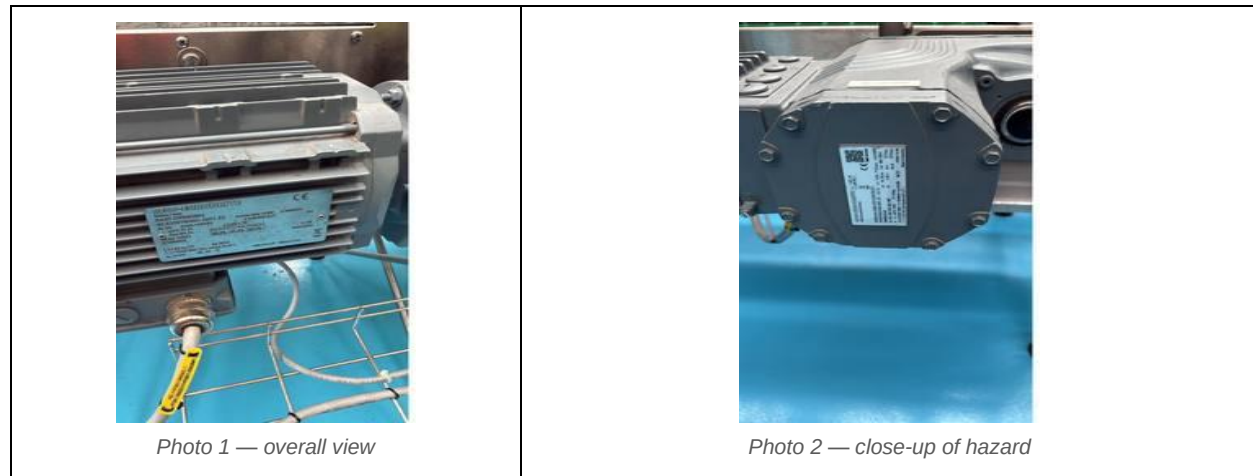


Figure HZ-04: Drive motor and timing belt

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Fixed guarding on belt, chain and shaft drives	Existing
Guard	G001	Fixed guard per ISO 14120 on drive assemblies	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	LOTO with standstill verification	Critical
Admin	A006	Periodic inspection of drive guards	Existing

► ACTION REQUIRED

- [A013] LOTO with standstill verification — Critical

Hazard Zone HZ-05: Pneumatic System

Hazard Source:	Pneumatic valve islands, cylinders and actuators	Consequence:	Impact / Crushing / Whip
Risk IDs:	R021-R025	Tasks:	T02,T05,T10,T11
Risk Parameters:	S2 / F1 / P1 => PLr c		

Operating pressure 4-6 bar. Include pneumatic isolation and bleed-down in LOTO — stored pneumatic energy can move cylinders after electrical isolation.

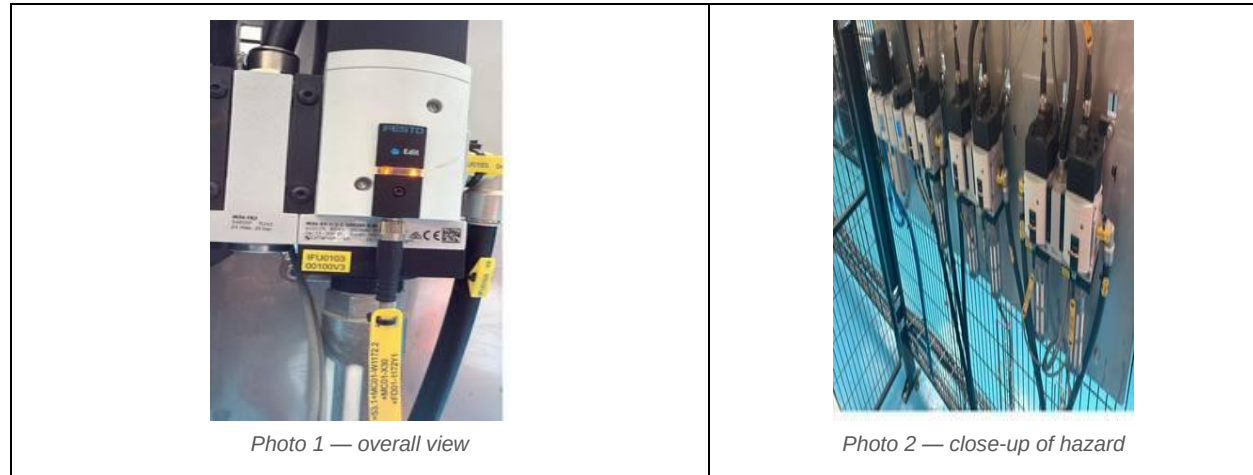


Figure HZ-05: Pneumatic valve island and regulators

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Shrouded pneumatic connections	Existing
Device	T016	Pneumatic shut-off and exhaust valve	Existing
Device	T011	E-Stop also isolates pneumatic energy	Existing
Admin	A013	LOTO with pneumatic isolation and bleed-down	Critical

► ACTION REQUIRED

- [A013] LOTO with pneumatic isolation and bleed-down — Critical

Hazard Zone HZ-06: Control Panel & Live Terminals

Hazard Source:	Main electrical panel with surge protection	Consequence:	Electrocution
Risk IDs:	R026-R030	Tasks:	T02,T05,T06,T09,T11
Risk Parameters:	S2 / F1 / P2 => NA		

400V terminals in the main panel. Important note: warning labels in the panel confirm some circuits remain live with the main isolator off. The machine-specific LOTO procedure must call these out by tag.

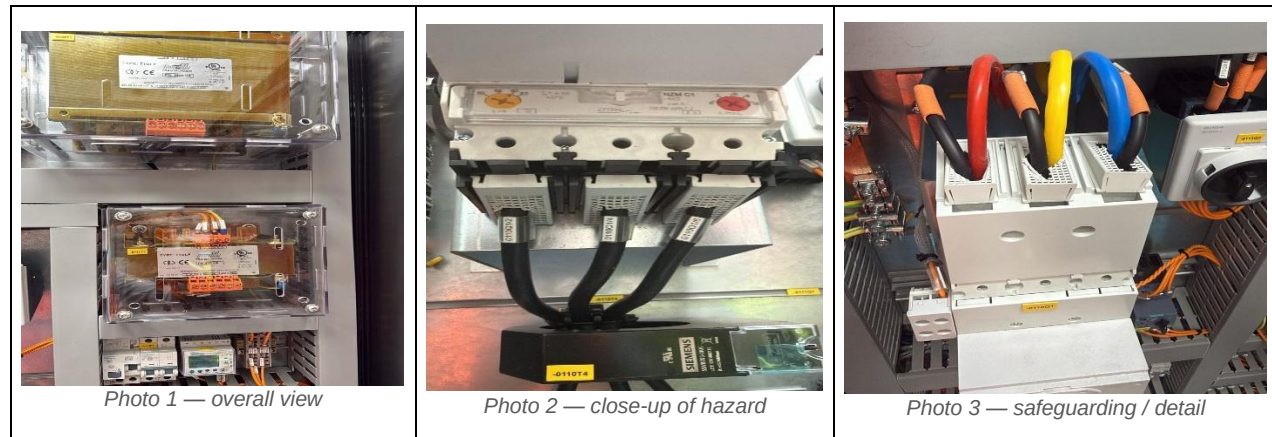


Figure HZ-06: OCME packer main electrical panel

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	IP-rated enclosures	Existing
Guard	G001	Fixed guard on control panel	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Device	T021	Dual earthing PE protection	Existing
Device	T019	Surge protection devices	Existing
Device	T011	E-Stop push buttons	Existing
Admin	A013	Machine-specific LOTO – flag permanently live circuits	Critical
Admin	A004	Live work by qualified electricians only	Recommended

► ACTION REQUIRED

- **[A013]** Machine-specific LOTO – flag permanently live circuits — *Critical*
- **[A004]** Live work by qualified electricians only — *Recommended*

Hazard Zone HZ-07: Electrical Earthing & Over-Voltage

Hazard Source:	Machine frame earthing and over-voltage protection	Consequence:	Electrocution / Shock
Risk IDs:	R031-R036	Tasks:	T02,T05,T06,T10,T11,T13
Risk Parameters:	S2 / F2 / P1 => PLr d		

Surge protection in the panel confirms over-voltage awareness. Two PE conductors per IEC 60204-1, annual continuity testing.



Photo 1 — overall view

Figure HZ-07: Earthing and bus bar arrangement

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Equipotential bonding of all metalwork	Existing at motors – body earthing to be verified
Device	T021	Dual earthing PE protection	Existing
Device	T019	Surge protection devices	Existing
Device	T008	Main isolator (lockable OFF)	Existing
Admin	A013	LOTO before any electrical work	Critical
Admin	A006	Annual earth continuity / insulation resistance testing	Critical

► ACTION REQUIRED

- [A013] LOTO before any electrical work — Critical
- [A006] Annual earth continuity / insulation resistance testing — Critical

Hazard Zone HZ-08: General Moving Parts – Awareness

Hazard Source:	Multiple simultaneously moving assemblies	Consequence:	Loss of Awareness / Ergonomic
Risk IDs:	R037-R039	Tasks:	T01,T11,T13
Risk Parameters:	S1 / F2 / P2 => PLr c		

Carriage, sealing bar, conveyors, drives — habituation risk over time. Film roll changeover and jam clearing add ergonomic strain.



Figure HZ-08: Film carriage area

Risk Reduction Measures:

Category	Ref.	Measure	Status
Inherent Design	D001	Transparent guard panels where practical	Existing
Device	T001	Audible/visible warning before automatic start	Existing
Admin	A001	Stack lights/beacons during operation	Existing
Admin	A002	Awareness signs at access points	Existing
Admin	A019	Mandatory PPE	Existing

F.9 KORE Compliance – OCME Packer

KORE compliance status for the OCME Packer. Items marked Action Req. require client follow-up.

KORE Requirement Area	Report Finding	Status	Section Ref.
Hazard ID & Risk Assessment	Full ISO 12100 assessment; 39+ hazard-task combinations.	Compliant	Sec. F.7
Machine Guarding	Perimeter guards, interlocked doors, light curtains, mode selectors adequate.	Compliant	F.7
LOTO	Electrical (with permanently live circuits flagged), pneumatic, thermal isolation.	Action Req.	HZ-03, HZ-06
Electrical Safety	Main isolator, E-stops, dual earthing PE, surge protection. Permanently live circuits flagged.	Action Req.	HZ-06, HZ-07
Emergency Stop	Coverage adequate.	Compliant	F.7
Training & Competency	Film threading, sealing bar, LOTO SOPs to be issued.	Action Req.	F.4, F.5
Inspection & PM	Periodic inspection programme to be set up.	Action Req.	F.7
Contractor & Visitor Safety	Visitor exposure assessed.	Compliant	HZ-01
PPE	Safety footwear, ear protection, heat-resistant gloves near sealing bar.	Compliant	F.7
Management of Change	Modifications require re-assessment.	Advisory	-

G. Authorisation & Sign-off

Prepared by – Independent SME on Machine Safety

Name	Muthu Vignesh R
Title / Qualification	CMSE® - Certified Machine Safety Expert
Date	13/05/2026
Signature	

Reviewed by

Name	
Title	
Date	13/05/2026
Signature	

Acknowledged by – Coca-Cola AI Ahlia Beverages Company LLC

Name	
Title	
Date	
Signature	

H. Cross-Reference Appendix — Tasks & Risk Reduction Measures

This appendix consolidates the Task codes (T-codes) and Risk Reduction Measure codes used throughout this report so readers can correlate references back to their definitions. Use this section as a quick lookup when reading hazard zone tables in any machine section.

H.1 How the Codes Work

Two distinct numbering schemes are used in this report. They look similar but they're not the same thing — pay attention to context.

Task codes (T01, T02, ...)

Task codes are scoped per machine. The Depalletizer's T05 is not the same task as the Filler's T05. Each machine section opens with its own task table (subsection X.6). The T-code numbering is consistent within a machine but resets between machines.

Risk Reduction Measure codes (D-, G-, T-, A- prefixed)

Measure codes are globally consistent across the entire report. D001, G001, T010, A013 mean the same thing wherever they appear. The 'T' prefix in measure codes denotes safety Devices (e.g. T010 = light curtain) and is unrelated to Task T-codes.

Quick rule: if the code stands alone (T05, T11) it's a Task — check the machine's own task table. If the code carries a category label or appears in a hazard zone measures table (Inherent Design / Guard / Device / Admin), it's a Risk Reduction Measure — check the master index in section H.3.

H.2 Task Codes — Per Machine

Tasks are listed in the order they appear in each machine's section. Common patterns are shared, but specifics vary by machine. Differences from the common pattern are highlighted.

A. Conveyor Zones

Conveyor zones do not have their own dedicated task list. They reference the task codes of the connected upstream and downstream machines. The tasks most relevant to conveyor work are T01 (monitoring), T02 (sensor adjustment), T05 (jam clearing), T11 (drive/pneumatic maintenance), and T12 (cleaning inside safeguarded space) from any machine.

B. Depalletizer (full task list)

#	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor / device adjustment	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans / pallets	Operator	Stopped	As needed
T06	Manual jog of pusher / arm	Operator	Manual	As needed
T07	Inspection of arm and clamp areas	Operator	Auto	Continuous
T08	Pallet stack management	Operator	Auto	Continuous
T09	HMI parameter changes	Maintenance	Manual	Periodic

T10	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T11	Pneumatic system maintenance	Maintenance	LOTO	Scheduled
T12	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T13	Format / changeover	Maintenance	Manual	Periodic

C. Palletizer — differences from Depalletizer

Tasks T01-T04, T09, T10, T12, T13 are identical to the Depalletizer. The following tasks differ:

#	Task (varies from common pattern)	Persons	Mode	Frequency
T05	Clearing jammed packs / pads	Operator	Stopped	As needed
T06	Manual jog of arms	Operator	Manual	As needed
T07	Inspection of arm and pad areas	Operator	Auto	Continuous
T08	Pallet / pad replenishment	Operator	Auto	Continuous
T11	Pneumatic / drive maintenance	Maintenance	LOTO	Scheduled

D. Filler (full task list — 11 tasks)

Filler task list is shorter than the Depalletizer / Palletizer pattern. T09-T13 do not align with the other machines — see numbering below.

#	Task	Persons	Mode	Frequency
T01	Monitoring production	Operator	Auto	Continuous
T02	Sensor adjustment / fault-finding	Maintenance	Manual	As needed
T03	Housekeeping / cleaning	Housekeeping	Stopped	Per shift
T04	Presence near machine	Visitors, Operators	Auto	Continuous
T05	Clearing jammed cans	Operator	Stopped	As needed
T06	Manual jog of carousel / starwheel	Operator	Manual	As needed
T07	Inspection of carousel	Operator	Auto	Continuous
T08	HMI parameter changes	Maintenance	Manual	Periodic
T09	PM / breakdown maintenance	Maintenance	LOTO	Scheduled
T10	Cleaning inside safeguarded space	Housekeeping, Operator	LOTO	Per shift
T11	Setting / changeover	Maintenance	Manual	Periodic

E. Can Seamer — differences from Depalletizer

Tasks T01, T03, T04, T09, T10, T12 are identical to the Depalletizer. The following tasks differ:

#	Task (varies from common pattern)	Persons	Mode	Frequency
T02	Sensor adjustment / fault-finding	Maintenance	Manual	As needed
T05	Clearing jammed cans / lids	Operator	Stopped	As needed

T06	Manual jog of seaming turret	Operator	Manual	As needed
T07	Inspection of seaming turret	Operator	Auto	Continuous
T08	Lid replenishment	Operator	Auto	Continuous
T11	Seaming roll / chuck changeover	Maintenance	LOTO	Periodic
T13	Setting / changeover	Maintenance	Manual	Periodic

F. OCME Packer — differences from Depalletizer

Tasks T01-T04, T09, T10, T12, T13 follow the common pattern. The following tasks differ:

#	Task (varies from common pattern)	Persons	Mode	Frequency
T05	Clearing jammed cans / film	Operator	Stopped	As needed
T06	Manual jog of carriage / conveyors	Operator	Manual	As needed
T07	Film roll replenishment / threading	Operator	Manual / Stopped	As needed
T08	Sealing bar adjustment / replacement	Maintenance	Manual / LOTO	Periodic
T11	Drive belt / gearbox maintenance	Maintenance	LOTO	Scheduled

H.3 Risk Reduction Measure Codes — Master Index

These codes are globally consistent across all six sections of this report. Where a code is referenced in any hazard zone table, the meaning is as follows.

H.3.1 Inherent Design Measures (D-codes)

Inherent safe design — built into the machine geometry and construction. Cannot be defeated by the operator.

Code	Category	Standard Meaning
D001	Inherent Design	Inherent safe design measures: rounded edges, minimum gaps per ISO 13854, mechanical stops, transparent guard panels, equipotential bonding of all metalwork, IP-rated enclosures, surface temperature limits per ISO 13732-1, shrouded pneumatic connections with secured hose routing, side guards on belt return rollers, safe geometry on lid feed components, mechanical blocks for arm at top of stroke.

H.3.2 Guards (G-codes)

Physical barriers between personnel and hazards. Per ISO 14120 (general guard requirements) and ISO 14119 (interlocking devices).

Code	Category	Standard Meaning
G001	Fixed Guard	Fixed guards per ISO 14120 — applied to belt and chain drives, conveyor drive heads, motor couplings, control panel enclosures. Guard fixings require tools for removal.
G003	Perimeter Guard	Perimeter guard with interlocked access doors. Prevents access to the hazardous operating envelope during automatic operation.

G004	Movable Guard	Movable (hinged) access door fitted with interlock and panic release. Used at routine access points where guards open for production-related tasks.
-------------	----------------------	---

H.3.3 Safety Devices (T-codes — Devices)

Active protective devices — electrical, electronic, or fluid-power components that detect or respond to hazards.

Note: These T-codes refer to Devices, not Tasks. The 'T' prefix here is the standard ISO 13849-1 device-category nomenclature used in this report's hazard zone tables. They are unrelated to the Task T-codes in section H.2.

Code	Category	Standard Meaning
T001	Warning	Audible / visible warning signal before automatic start-up.
T005	Reset	Reset button (blue) with indicator, positioned inside the safeguarded space. PLr e — typically required after light curtain or interlock activation.
T008	Isolator	Local / main electrical isolator with lockable OFF position. Required at each conveyor section and on the main panel of each machine. Per IEC 60204-1.
T009	Muting	Cross-beam muting per IEC 62046 — temporarily bypasses light curtain for legitimate material passage while maintaining detection of personnel.
T010	Light Curtain	Safety light curtain positioned per ISO 13855 (approach speeds). PLr e. Used at infeed/discharge openings of guarded zones.
T011	E-Stop	Emergency stop push buttons / pull-cords per ISO 13850. PLr d. Required within reach from any operator working position.
T012	Cascade Stop	Cascade stop logic between linked conveyor sections / pull-cord with double-action latching. Ensures upstream and downstream stop in synchrony.
T014	Interlock	Interlock with panic release per ISO 14119. PLr d. Applied to all access doors in perimeter guards.
T015	Mode Selector	Mode selector key switch (selectable between Auto, Manual, Service modes). PLr c. Ensures hazardous functions are disabled in non-production modes.
T016	Pneumatic Isolation	Pneumatic shut-off and exhaust valve per EN ISO 4414. Isolates and bleeds stored pneumatic energy as part of LOTO.
T019	Over-Voltage	Over-voltage / surge protection device installed in the main electrical panel.
T021	PE Earthing	Dual earthing — two independent protective conductors (PE) per IEC 60204-1 Clause 8.2.3.

H.3.4 Administrative Measures (A-codes)

Procedural and behavioural controls — the last line of defence in the hierarchy of risk reduction (per ISO 12100).

Code	Category	Standard Meaning
A001	Visual Indication	Stack lights / beacons during operation. Visible indication of machine status from any operator position.
A002	Warning Signs	Warning signs at access points: electrical hazard, hot surface, E-stop locations, machine area awareness, pusher arm access.

A004	SOP / Qualification	Qualified persons only for specific work (e.g. live electrical work). Written SOPs required for seaming roll/chuck adjustment, changeover, jam clearing near pneumatic actuators.
A006	Periodic Inspection	Periodic inspection programmes — annual earth continuity and insulation resistance testing, quarterly E-stop functional testing, periodic inspection of drive guards / belt tracking / idler condition / mechanical blocks / interlocks.
A007	Access Control	Measures to prevent unauthorised access to safeguarded zones.
A013	LOTO	Lockout / Tagout procedures. Machine-specific procedures required covering electrical isolation, pneumatic isolation and bleed-down, thermal cooldown, mechanical block engagement, standstill verification. Permanently live circuits (where applicable) must be identified in the procedure.
A019	PPE	Mandatory PPE in machine area — safety footwear, hand protection, heat-resistant gloves near sealing bar.

H.4 Risk ID Scoping

Risk IDs (R001, R002, ...) are also scoped per machine. The Depalletizer's R001–R005 are different risks from the Filler's R001–R005. Always read Risk IDs in the context of the machine section they appear in.

Globally unique codes in this report: D001, G001–G004, T001–T021 (devices), A001–A019 (administrative). All others (Risk IDs, Task T-codes) are scoped per machine.